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An Introduction to Module Theory

Ibrahim Assem and Flávio U. Coelho

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Great Clarendon Street, Oxford, OX2 6DP,
United Kingdom

Oxford University Press is a department of the University of Oxford.
It furthers the University's objective of excellence in research, scholarship,
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Oxford University Press in the UK and in certain other countries

Published in the United States of America by Oxford University Press
198 Madison Avenue, New York, NY 10016, United States of America

British Library Cataloguing in Publication Data
Data available

Library of Congress Control Number: 2024939638

ISBN 9780198904908
ISBN 9780198904915 (pbk.)

DOI: 10.1093/9780198904939.001.0001

Printed and bound by
CPI Group (UK) Ltd, Croydon, CR0 4YY

Dedicated to Sonia and Marcia

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Introduction

This book is a byproduct of graduate courses taught by the authors in their respective universities over the last thirty years. During this period, the area it addresses—Module theory—has experienced deep changes, to which teaching had to adapt. Module theory is one of the fundamental areas of algebra and has important applications in other parts of mathematics—such as in Representation Theory (of groups and algebras), Combinatorics, Geometry, Algebraic Topology, Mathematical Physics and Functional Analysis. It is being taught at the Master level to students in algebra, and also to students who need algebraic tools in their work. We believe our book to be an accessible and reasonably selfcontained account of Module theory, that can be used as a textbook for graduate students, advanced undergraduate students and also for selfstudying. In fact, we only assume that the reader knows some linear algebra and ring theory, at the level normally acquired in standard undergraduate courses of most universities.

Module theory can be thought of as an advanced form of linear algebra. Indeed, the definition of a module is the same as that of a vector space, except that the scalars, instead of being taken in a field, are taken in a ring, or, equivalently, in an algebra. In other words, scalars do not necessarily have multiplicative inverses. This makes their study very different from that of vector spaces—perhaps the most striking difference comes from the fact that, usually, a module does not have a basis. Another one is that, because our rings (or algebras) are not assumed to be commutative, one has to consider the two possibilities for multiplying scalars, namely, on the right or on the left. These, and other differences, make the setting of modules much richer than that of vector spaces and call for new methods and tools in order to study this algebraic structure.

Module theory is a standard area but it has undergone profound changes in recent years, the most evident one being the new emphasis on categorical and homological techniques. Our approach stresses the use of both from the start. Category theory, originally introduced as an elegant unifying language, furnishes a deep conceptual insight into the theory. It provides a new light in which to look at modules and, therefore, new research problems. From the practical point of view, category theory forces us to concentrate on maps rather than on sets and their elements, and this allows us to present simpler and shorter proofs of classical results. We do not assume any previous knowledge of category theory. Instead, we provide all relevant definitions and results. Similarly, homological algebra, which takes its origins in algebraic topology, is used to provide invariants for modules. Accordingly, our book features a selfcontained introduction to homological methods and their applications to module theory.

Our treatment is based on the strong conviction that the top priority should be given to clarity and intelligibility of the text. Thus we tried to provide the reader with

An Introduction to Module Theory. Ibrahim Assem and Flávio U. Coelho, Oxford University Press.
 DOI: 10.1093/9780198904939.003.0001

as many pictures and examples as possible. Mathematicians do a lot of examples and draw a lot of pictures on the blackboard to help visualising concepts and thus develop their students' intuition, but these pictures seldom make it into the books. We made it a point to include as many pictures as possible, explaining their advantages as well as their limitations. We included a large number of examples of all kinds worked out in detail. We focused on showing the student how to construct and analyse examples by himself, in a systematic way, leading to the proof of a statement or a counterexample to an incorrect one. Besides taking our examples from classical areas, such as abelian groups or modules over matrix rings, we introduce representations of quivers, by now a well-developed and established research area, which provides us with an inexhaustible source of examples, elementary enough for any beginner to handle (and to construct by himself). Finally, we try to motivate the introduction of each new concept by explaining, in a few words, why it is needed. Likewise, we present in several places the rationale for new results and the strategy followed in order to prove them. We did not hesitate to repeat ourselves when it seemed to us to enhance our exposition. As a consequence, the pace of our presentation is deliberately pedestrian and we try to give as many details as possible, especially in the proofs of the main results. We have included a large number of exercises of all levels of difficulty, both computational and theoretical, and provided hints for some of them.

The choice of the material presented here and our approach to it are consequences both of our teaching experience and our mathematical itinerary. The contents have nothing original: the concepts and results presented here have appeared in different forms in many other books or research papers. We acknowledge our debt to the texts cited in the bibliography, and to so many others we did not specifically mention. This book is conceived as a textbook for a beginner Master Student in Mathematics. Due to this constraint, we could not be encyclopedic in this work. Perhaps the most obvious omission is the study of limits and colimits in module categories, leading to the description of infinitely generated modules. We apologise to the reader for this and any other omission.

We explain the contents of the book. Chapter I recalls the necessary facts from undergraduate ring theory and presents the equivalent concept of an algebra. We start working with modules in Chapter II, treating them both from a classical perspective and from a categorical one. Chapters III and IV are devoted to the rudiments of category theory and direct applications in module theory. In Chapter V, we pause from theory to present a classical topic: the classification of finitely generated modules over a principal ideal domain, with its straightforward application to the Jordan form of a matrix. Chapter VI concentrates on the study of two functors, namely, the Hom and tensor functors, which allow to pass from one module category to another and to introduce projective, injective and flat modules. Chapters VII and VIII contain the classical results of module theory, such as the Jordan–Hölder, Wedderburn–Artin, Hopkins–Levitsky theorems, and finally, the Krull–Schmidt Unique Decomposition theorem, which shows how a module can be built starting from ones that cannot be decomposed into direct sums of smaller ones. In Chapter IX, we describe finitely generated projective modules over Artinian algebras, we prove the Morita and Eilenberg–Watts theorems, then return to representations of quivers, culminating in the computation of indecomposable finitely generated modules over the so-called

Nakayama algebras. Chapter X is meant as an introduction to the main tools of homological algebra, the (co)homology sequences and derived functors. These are applied in Chapter XI, where we present extension and torsion modules and the resulting functors. We conclude in Chapter XII by discussing homological dimensions of modules and algebras and showing how they can be used in the study of hereditary and selfinjective algebras.

This book has developed from graduate courses given at the universities of Sherbrooke and São Paulo. It is a pleasure for us to acknowledge our debt to our students and all the colleagues with whom we discussed the contents of our courses. Their questions, remarks and criticisms gave us an invaluable feedback. We thank in particular Shiping Liu, Marco Armenta and Juan Carlos Bustamante for their useful suggestions.

We also warmly thank Jean-Philippe Morin and Souheila Hassoun for their precious help in getting our manuscript into shape. We are also grateful to Dan Taber, Giulia Lipparini as well as the whole OUP team for their help during its preparation.

The authors gratefully acknowledge financial support from CNPq and FAPESP, Brazil, as well as NSERC, Canada.

Sherbrooke and São Paulo,
January 2024

Chapter I
Rings and Algebras

I.1 Introduction

The subject of this book is module theory. The easiest way to think of a module is as 'a vector space over a ring' or, equivalently, 'over an algebra'. More precisely, a *module* is defined by the same axioms as a vector space but the scalars, instead of being taken inside a field, are taken inside a ring, or an algebra. However, different techniques are needed to deal with modules. Because nonzero scalars do not usually have inverses, many basic properties of vector spaces do not hold true for modules in general. For instance, modules do not necessarily have bases. In this first chapter, we recall the basic properties of rings and introduce the equivalent notion of algebra, which is technically easier and more fruitful. We also present several examples.

I.2 Rings and modules

I.2.1 Rings

In elementary mathematics, most sets under consideration, such as integers, real numbers, polynomials and matrices, have (at least) two operations, 'addition' and 'multiplication', and many properties of these sets arise from the interplay between these operations. If X is a set, an ***operation*** or ***internal operation*** in X is a map from $X \times X$ to X. That is, it assigns to each pair of elements of X a uniquely determined element of X, called ***result*** of the operation.

Typical operations are the usual addition and multiplication of integers, real numbers, polynomials or matrices. There are differences between properties of these operations in these sets. For instance, multiplication of integers, real numbers or polynomials is commutative, but not multiplication of matrices. Further, any nonzero real number has a multiplicative inverse which belongs to the same set of nonzero real numbers, but this property does not hold true for integers, polynomials or matrices. Now, the set of real numbers is a typical example of a field, as defined in linear algebra courses. The other three sets mentioned satisfy all axioms of a field, except perhaps commutativity of multiplication, and existence of multiplicative inverses. Removing the latter two from the axioms of a field, we get the following definition.

I.2.1 Definition A ***ring*** $(K, +, \cdot, 1)$, or simply K, is a quadruple, where K is a set such that:

(A) There is an operation $+$ on K, called ***addition***, which associates to each pair $(\alpha, \beta) \in K \times K$ its ***sum*** $\alpha + \beta \in K$, and which makes the pair $(K, +)$ an ***abelian group***, that is, it satisfies the following axioms:

An Introduction to Module Theory. Ibrahim Assem and Flávio U. Coelho, Oxford University Press.
 DOI: 10.1093/9780198904939.003.0002

- (A1) It is commutative, that is, $\alpha + \beta = \beta + \alpha$, for any $\alpha, \beta \in K$.
- (A2) It is associative, that is, $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$, for any $\alpha, \beta, \gamma \in K$.
- (A3) There exists an element $0 \in K$, called the ***zero***, such that $\alpha + 0 = 0 + \alpha = \alpha$, for any $\alpha \in K$.
- (A4) To any $\alpha \in K$ is associated an element $(-\alpha) \in K$, called its ***negative***, or ***opposite***, such that $\alpha + (-\alpha) = (-\alpha) + \alpha = 0$.

(B) There is an operation $\cdot$ on K, called ***multiplication***, which associates to each pair $(\alpha, \beta) \in K \times K$ its ***product*** $\alpha \cdot \beta \in K$, also denoted simply by $\alpha\beta$, and an element 1 of K, which make the triple $(K, \cdot, 1)$ a ***monoid***, that is, it satisfies the following axioms:

- (B1) It is associative, that is, $\alpha(\beta\gamma) = (\alpha\beta)\gamma$, for any $\alpha, \beta, \gamma \in K$.
- (B2) The element $1 \in K$, called the ***identity***, or ***one***, is such that $\alpha 1 = 1\alpha = \alpha$, for any $\alpha \in K$.

(C) These operations are compatible, in the sense that multiplication is ***left and right distributive*** over addition:

- (C1) $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$, for any $\alpha, \beta, \gamma \in K$.
- (C2) $(\beta + \gamma)\alpha = \beta\alpha + \gamma\alpha$, for any $\alpha, \beta, \gamma \in K$.

Some authors consider rings without identity elements, that is, abelian groups with a multiplication satisfying axioms (B1), (C1) and (C2) (but not necessarily (B2)). In this text, we always consider rings with identities, sometimes called ***unitary rings***. This assumption is usual when one deals with modules, and can be made without loss of generality, as explained after Definition I.3.1.

While we required addition of a ring to be commutative, we made no such assumption on multiplication. This is an additional axiom that a ring may or may not satisfy. A ring K is called ***commutative*** if $\alpha\beta = \beta\alpha$, for any $\alpha, \beta \in K$. In a commutative ring, the left distributivity axiom (C1) is equivalent to the right (C2).

I.2.2 Examples.

(a) One may ask what is the 'smallest' possible ring. Because of Definition I.2.1, any ring contains 0 and 1, so no ring is empty. But nothing says that 0 and 1 should be distinct. Indeed, let $K = \{0\}$ be a one-element set, and define operations by setting $0 + 0 = 0$ and $0 \cdot 0 = 0$. Then K becomes a (commutative) ring, called the ***zero*** or ***trivial*** ring, and denoted by $K = 0$.

(b) The sets $\mathbb{Z}$ of integers, $\mathbb{Q}$ of rationals, $\mathbb{R}$ of reals and $\mathbb{C}$ of complexes, with their usual addition and multiplication, are commutative rings.

(c) For any integer $m \geq 2$, the set $\mathbb{Z}/m\mathbb{Z}$ of integers modulo m is a commutative ring. Indeed, let $\bar{a} = a + m\mathbb{Z}$ denote the residual class of an integer a. It is easily verified that the addition and multiplication given respectively by $\bar{a} + \bar{b} = \overline{a + b}$ and $\bar{a} \cdot \bar{b} = \overline{ab}$, for $a, b \in \mathbb{Z}$, are unambiguously defined operations which satisfy the required axioms.

(d) Let K be a ring. The set $K[t]$ of polynomials in one indeterminate t with coefficients in K is a ring under ordinary addition and multiplication of polynomials. The ring $K[t]$ is commutative if (and only if) K is commutative.

(e) Let K be a ring and n a positive integer. The set $M_n(K)$ of $n \times n$-matrices with coefficients in K is a ring under ordinary addition and multiplication of matrices. If $n > 1$ and $1 \neq 0$ in K, then $M_n(K)$ is not commutative even if K is commutative. For instance, the matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ do not commute.

(f) Let K be a ring. The ***opposite ring*** K^{op} has the same elements and the same addition as K, so that $K^{\text{op}} = K$, *as abelian groups*, but multiplication in K^{op} is defined by $\alpha \cdot \beta = \beta\alpha$, for $\alpha, \beta \in K$, where the product on the right-hand side is the product in K. Thus, a ring K is commutative if and only if $K^{\text{op}} = K$, *as rings*.

Rings satisfy most arithmetic properties of integers, as is shown by the following proposition.

I.2.3 Proposition. *Let K be a ring, then:*
(a) *The zero element of K is unique.*
(b) *The identity element of K is unique.*
(c) *The negative of any $\alpha \in K$ is unique.*
(d) *For any $\alpha \in K$, we have $\alpha \cdot 0 = 0 \cdot \alpha = 0$.*
(e) *For any $\alpha, \beta \in K$, we have $(-\alpha)\beta = \alpha(-\beta) = -(\alpha\beta)$.*
(f) *For any $\alpha, \beta_1, \ldots, \beta_m \in K$, we have $\alpha(\sum_{i=1}^m \beta_i) = \sum_{i=1}^m (\alpha\beta_i)$.*
(g) *For any $\alpha_1, \ldots, \alpha_n, \beta \in K$, we have $(\sum_{j=1}^n \alpha_j)\beta = \sum_{j=1}^n (\alpha_j\beta)$.*

Proof. (a) If both $0, 0'$ satisfy axiom (A3) then, because 0 is a zero, we have $0+0' = 0'$. But $0'$ is also a zero, so $0 + 0' = 0$. Hence, $0 = 0'$.
(b) This is similar to (a) and left to the reader.
(c) If both $(-\alpha)$ and $(*\alpha)$ satisfy (A4), then, thanks to (A2),

$$(-\alpha) = (-\alpha) + 0 = (-\alpha) + (\alpha + (*\alpha)) = ((-\alpha) + \alpha) + (*\alpha) = 0 + (*\alpha) = (*\alpha).$$

(d) Let $\alpha \in K$. Then $0 + 0 = 0$ gives, thanks to (C1),

$$\alpha \cdot 0 = \alpha \cdot (0 + 0) = \alpha \cdot 0 + \alpha \cdot 0.$$

Subtracting $\alpha \cdot 0$ from both sides, that is, adding $-(\alpha \cdot 0)$ to both sides, we get $\alpha \cdot 0 = 0$. Similarly, $0 \cdot \alpha = 0$.
(e) Let $\alpha, \beta \in K$. Because of (C2) and (d) above, we have:

$$(-\alpha)\beta + \alpha\beta = ((-\alpha) + \alpha) \cdot \beta = 0 \cdot \beta = 0.$$

Then (c) above gives $(-\alpha)\beta = -(\alpha\beta)$. Similarly, $\alpha(-\beta) = -(\alpha\beta)$.
(f) This is proved by induction on m. If $m = 1$, there is nothing to prove. If $m > 1$, then, using the induction hypothesis and (C1), we have:

$$\begin{aligned}\alpha\left(\sum_{i=1}^{m}\beta_i\right) &= \alpha\left(\left(\sum_{i=1}^{m-1}\beta_i\right)+\beta_m\right) = \alpha\left(\sum_{i=1}^{m-1}\beta_i\right)+\alpha\beta_m \\ &= \sum_{i=1}^{m-1}(\alpha\beta_i)+\alpha\beta_m = \sum_{i=1}^{m}\alpha\beta_i.\end{aligned}$$

(g) This is similar to (f) and left to the reader. □

Because K is an abelian group, we also have $-(-\alpha) = \alpha$ and $-(\alpha+\beta) = (-\alpha)+(-\beta)$, for $\alpha, \beta \in K$.

A nice consequence of (d) is that a ring equals the zero ring if and only if its identity equals zero. Clearly, $K = 0$ implies $1 = 0$. Conversely, if $1 = 0$ in K, then, for any $\alpha \in K$, we have $\alpha = \alpha \cdot 1 = \alpha \cdot 0 = 0$, so $K = 0$.

When we wrote that rings satisfy 'most' arithmetic properties of integers, we meant that they usually do not satisfy them all. For instance, the product of two nonzero integers is nonzero, but in the ring $\mathbb{Z}/6\mathbb{Z}$, the product of the nonzero elements $\bar{2}$ and $\bar{3}$ is $\bar{2}\cdot\bar{3} = \bar{6} = \bar{0}$. Such elements are called zero divisors.

I.2.4 Definition. Let K be a nonzero commutative ring.

(a) A nonzero element $\alpha \in K$ is a ***zero divisor*** if there exists a nonzero element $\beta \in K$ such that $\alpha\beta = 0$.

(b) The ring K is an ***integral domain*** if it contains no zero divisor.

Thus, a nonzero commutative ring K is an integral domain if and only if it satisfies one of the equivalent conditions:

(i) If $\alpha, \beta \in K$ are such that $\alpha\beta = 0$, then $\alpha = 0$ or $\beta = 0$.

(ii) If $\alpha, \beta \in K$ are such that $\alpha\beta = 0$ and $\alpha \neq 0$, then $\beta = 0$.

(iii) If $\alpha, \beta \in K$ are such that $\alpha \neq 0$ and $\beta \neq 0$, then $\alpha\beta \neq 0$.

A typical example of an integral domain is the set $\mathbb{Z}$ of integers. If K is an integral domain, then the ring of polynomials $K[t]$ is also an integral domain, because the product of the (nonzero) leading coefficients of two nonzero polynomials is itself nonzero.

In noncommutative rings, one distinguishes left and right zero divisors, but we shall not need that.

A nonzero commutative ring is an integral domain if and only if it satisfies the so-called ***cancellation property***.

I.2.5 Lemma. *A nonzero commutative ring K is an integral domain if and only if, for any $\alpha, \beta, \gamma \in K$ such that $\alpha \neq 0$ and $\alpha\beta = \alpha\gamma$, we have $\beta = \gamma$.*

Proof. *Necessity.* From $\alpha\beta = \alpha\gamma$ we infer that $\alpha(\beta-\gamma) = 0$. Because K is integral and $\alpha \neq 0$, this gives $\beta - \gamma = 0$, hence $\beta = \gamma$.

Sufficiency. If $\alpha \neq 0$ and $\alpha\beta = 0$, then $\alpha\beta = \alpha 0$ yields $\beta = 0$, hence K is integral. □

We now consider multiplicative inverses in rings. Let K be a ring. An element $\alpha \in K$ is called ***invertible*** if there exists $\alpha^{-1} \in K$ such that $\alpha \cdot \alpha^{-1} = \alpha^{-1} \cdot \alpha = 1$. If such an element α^{-1} exists, then it is easily shown, as in Proposition I.2.3(c), that it is unique. It is called the ***inverse*** of α. As expected, if $\alpha, \beta \in K$, then $(\alpha^{-1})^{-1} = \alpha$ and $(\alpha\beta)^{-1} = \beta^{-1}\alpha^{-1}$ whenever α^{-1}, β^{-1} exist. In the ring $\mathbb{Z}$ of integers, only the elements 1 and -1 have multiplicative inverses. It follows from Proposition I.2.3(d) that, in a nonzero ring, the zero element is not invertible.

I.2.6 Definition. Let K be a nonzero ring. If any nonzero element of K is invertible, then K is a ***division ring*** or a ***skew field***. If, moreover, K is commutative, then it is a ***field***.

The reader is certainly familiar with several examples of fields, such as $\mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$. Before giving more examples, we observe the following.

I.2.7 Lemma. *Invertible elements in a commutative ring are not zero divisors. In particular, any field is an integral domain.*

Proof. Assume that $\alpha \neq 0$ and $\alpha\beta = 0$. If α^{-1} exists, then

$$\beta = 1 \cdot \beta = (\alpha^{-1}\alpha)\beta = \alpha^{-1}(\alpha\beta) = \alpha^{-1} \cdot 0 = 0.$$

This proves the first statement. The second is an obvious consequence. □

The converse of the lemma is false: $\mathbb{Z}$ is an integral domain but not a field. However, any *finite* integral domain is a field. For, let K be a finite integral domain and $\alpha \in K$ be nonzero. Define a map $f : K \to K$ by $x \mapsto \alpha x$, for $x \in K$. Because of Lemma I.2.5, f is injective. But K is a finite set, and a map from a finite set to itself is injective if and only if it is surjective. Hence, $f : K \to K$ is surjective. So there exists $x \in K$ such that $1 = f(x) = \alpha x$. Thus, α is invertible, as required.

I.2.8 Examples.

(a) The best known example of a noncommutative division ring is the set $\mathbb{H}$ of Hamilton's quaternions. A ***quaternion*** is a 2×2 matrix of the form

$$\begin{pmatrix} u & v \\ -\bar{v} & \bar{u} \end{pmatrix}$$

where u, v are complex numbers, and $\bar{u}, \bar{v}$ their respective conjugates. We leave to the reader the verification that $\mathbb{H}$ is a ring under addition and multiplication of matrices. We prove that $\mathbb{H}$ is a division ring. Let $\begin{pmatrix} u & v \\ -\bar{v} & \bar{u} \end{pmatrix} \in H$ be nonzero. Its determinant equals $u\bar{u} + v\bar{v} = |u|^2 + |v|^2 \neq 0$. Hence, this matrix has a multiplicative inverse, namely $\frac{1}{|u|^2+|v|^2}\begin{pmatrix} \bar{u} & -v \\ \bar{v} & u \end{pmatrix}$, which lies in $\mathbb{H}$. Therefore, $\mathbb{H}$ is a division ring. Multiplication in $\mathbb{H}$ is not commutative: for instance, the matrices $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, both lying in $\mathbb{H}$, do not commute. Therefore, $\mathbb{H}$ is a noncommutative division ring.

(b) Let $m \geq 2$ be an integer. The commutative ring $\mathbb{Z}/m\mathbb{Z}$ is a field if and only if m is prime. Indeed, assume that m is prime, and $\bar{a} \in \mathbb{Z}/m\mathbb{Z}$ is nonzero. Then there exists a unique $k \in \mathbb{Z}$ such that $0 < k < m$ and $\bar{k} = \bar{a}$ (actually, k is the remainder of the division of a by m). But then k and m are coprime, so, because of the Bézout–Bachet theorem, there exist $s, t \in \mathbb{Z}$ such that $sk + tm = 1$. Passing modulo m, this gives $\bar{s}\bar{k} = \bar{1}$, because $\bar{m} = \bar{0}$. Hence, $\bar{s} = \bar{k}^{-1} = \bar{a}^{-1}$ and $\bar{a}$ is invertible, as required.

Conversely, if m is not prime, there exist a, b such that $1 < a, b < m$ and $m = ab$. But then $\bar{a}, \bar{b}$ are nonzero, while $\bar{a}\bar{b} = \overline{ab} = \bar{m} = \bar{0}$. So, $\mathbb{Z}/m\mathbb{Z}$ has zero divisors. Because of Lemma I.2.7, it is not a field.

To end this subsection, we prove that any integral domain can be embedded in a field. The prototype of this construction is that of the field $\mathbb{Q}$ of rationals, starting from the integral domain $\mathbb{Z}$. If K, K' are rings, then a map $f : K \to K'$ is called a ***ring homomorphism*** provided it preserves addition, multiplication and the multiplicative identity, that is: $f(\alpha + \beta) = f(\alpha) + f(\beta), f(\alpha\beta) = f(\alpha)f(\beta)$, for any $\alpha, \beta \in K$ and $f(1) = 1$. A ***ring isomorphism*** is a ring homomorphism that is bijective.

I.2.9 Definition. Let K be a commutative ring. A field, or a division ring, K' is an ***extension*** of K if there exists an injective ring homomorphism from K to K'.

That is, K' is an extension of K if and only if it is a field, or a division ring, containing (an isomorphic image of) K as a subring. For instance, the inclusions $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ indicate three field extensions of the integers. The set of quaternions $\mathbb{H}$ is a (noncommutative) division ring extension of $\mathbb{C}$. The subfield $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ of $\mathbb{R}$ is a field extension of $\mathbb{Q}$.

Let K be an integral domain. Define an equivalence relation on the set $K \times K^* = \{(\alpha, \beta) : \alpha, \beta \in K, \beta \neq 0\}$ by setting:

$$(\alpha, \beta) \sim (\gamma, \delta) \text{ if and only if } \alpha \cdot \delta = \beta \cdot \gamma.$$

For instance, $(\alpha, \beta) \sim (0, \delta)$ if and only if $\alpha\delta = 0$. But $\delta \neq 0$ by definition, so $\alpha = 0$. Thus, $(\alpha, \beta) \sim (0, \delta)$ if and only if $\alpha = 0$.

We prove that $\sim$ is an equivalence. Reflexivity and symmetry are easy to verify. We prove transitivity. Assume that $(\alpha, \beta) \sim (\gamma, \delta)$ and $(\gamma, \delta) \sim (\lambda, \mu)$. If $\gamma = 0$, then the previous argument shows that $\alpha = \lambda = 0$ and there is nothing to show. So, suppose $\gamma \neq 0$. Then $\alpha\delta = \beta\gamma$ and $\gamma\mu = \delta\lambda$. Multiplying yields $\alpha\delta\gamma\mu = \beta\gamma\delta\lambda$. But $\delta \neq 0$, hence $\delta\gamma \neq 0$. Therefore, Lemma I.2.5 gives $\alpha\mu = \beta\lambda$, that is, $(\alpha, \beta) \sim (\lambda, \mu)$, as required.

We denote by α/β or $\frac{\alpha}{\beta}$ the equivalence class of a pair $(\alpha, \beta) \in K \times K^*$ and call it a ***fraction***. Let $Q(K) = (K \times K^*)/\sim$ be the quotient set. Define operations on $Q(K)$ by setting:

$$\frac{\alpha}{\beta} + \frac{\gamma}{\delta} = \frac{\alpha\delta + \beta\gamma}{\beta\delta}$$
$$\frac{\alpha}{\beta} \cdot \frac{\gamma}{\delta} = \frac{\alpha\gamma}{\beta\delta}$$

for $\alpha/\beta, \gamma/\delta \in Q(K)$.
We must prove that these operations are unambiguously defined. Assume that $(\alpha, \beta) \sim (\alpha', \beta')$, $(\gamma, \delta) \sim (\gamma', \delta')$. Then $\alpha\beta' = \alpha'\beta$ and $\gamma\delta' = \gamma'\delta$. Hence,

$$\begin{aligned}(\alpha\delta + \beta\gamma)\beta'\delta' &= \alpha\delta\beta'\delta' + \beta\gamma\beta'\delta' = \alpha\beta'\delta\delta' + \beta\beta'\gamma\delta' = \\ &= \alpha'\beta\delta\delta' + \beta\beta'\gamma'\delta = \alpha'\delta'\beta\delta + \beta'\gamma'\beta\delta = (\alpha'\delta' + \beta'\gamma')\beta\delta\end{aligned}$$

therefore $(\alpha\delta + \beta\gamma, \beta\delta) \sim (\alpha'\delta' + \beta'\gamma', \beta'\delta')$. Thus, addition of classes does not depend on the choice of the representatives. The proof for multiplication is similar (even easier).

I.2.10 Lemma. *With these operations, $Q(K)$ is a field extension of K.*

Proof. It consists in checking the axioms. The element $0/1$ is the zero, and $1/1$ the identity. The inverse of a nonzero element α/β is β/α (which makes sense because $\alpha \neq 0$). Finally, in order to consider K as embedded inside $Q(K)$, identify $\alpha \in K$ with the fraction $\alpha/1$. □

The field $Q(K)$ is called the ***field of fractions*** of the integral domain K. For instance, if K is a field, then the field of fractions of the integral domain $K[t]$ of polynomials in one indeterminate t is the field $K(t) = \{p/q : p, q \in K[t], q \neq 0\}$ of rational fractions.

It is well-known that a system of homogeneous linear equations with a nonzero rational solution also has a nonzero integral solution. This generalises to the following proposition.

I.2.11 Proposition. *Let K be an integral domain and $Q(K)$ its field of fractions. If the system of homogeneous linear equations*

$$\begin{cases} \alpha_{11}x_1 + \cdots + \alpha_{1n}x_n = 0 \\ \vdots \\ \alpha_{m1}x_1 + \cdots + \alpha_{mn}x_n = 0 \end{cases} \tag{*}$$

with coefficients $\alpha_{ij} \in K$ has a nonzero solution in $Q(K)^n$, then it has a nonzero solution in K^n.

Proof. Suppose $(\beta_1, \cdots, \beta_n) \in Q(K)^n$ is a nonzero solution of (*). For each i, write β_i as γ_i/δ_i, where $\gamma_i, \delta_i \in K$ and $\delta_i \neq 0$. Let $\delta = \delta_1 \cdots \delta_n$. Because $\delta_i \neq 0$, for each i, we have $\delta \neq 0$. Now, for each i, we have $\delta(\gamma_i/\delta_i) = \gamma_i'/1$, where $\gamma_i' = (\prod_{j\neq i} \delta_j)\gamma_i \in K$. Then

$$\delta(\beta_1, \cdots, \beta_n) = (\delta\beta_1, \cdots, \delta\beta_n) = (\gamma_1'/1, \cdots, \gamma_n'/1)$$

is also a solution of (*) which can be interpreted as $(\gamma_1', \cdots, \gamma_n') \in K^n$, as required. □

I.2.2 Modules

As we said, modules are 'vector spaces over rings' by which we mean that they have an addition and a scalar multiplication by elements of rings. Rings are not assumed commutative. Therefore, we must define two types of modules, one with the scalar multiplied on the right of the vector, and the other with the scalar multiplied on its left. These are respectively called right and left modules.

I.2.12 Definition. Let K be a ring. A ***right K-module*** $(M, +, \cdot)$, or simply M, is a triple, where M is a set such that:

(A) There is an operation $+$ on M, called ***addition***, which associates to each pair $(x, y) \in M \times M$ an element $x + y \in M$, called its ***sum***, and which makes the pair $(M, +)$ an abelian group, that is, it satisfies the following axioms:
  (A1) It is commutative, that is, $x + y = y + x$, for any $x, y \in M$.
  (A2) It is associative, that is, $x + (y + z) = (x + y) + z$, for any $x, y, z \in M$.
  (A3) There exists an element $0 \in M$, called the ***zero***, such that $x + 0 = 0 + x = x$, for any $x \in M$.
  (A4) To any element $x \in M$ is associated its ***negative***, or ***opposite***, $(-x) \in M$ such that $x + (-x) = (-x) + x = 0$.

(B) There is a map $M \times K \to M$, called ***external multiplication***, or ***right*** K-***action***, associating to each pair $(x, \alpha) \in M \times K$ an element $x \cdot \alpha$, or simply $x\alpha$, in M, called their ***product***, satisfying the following axioms:
  (B1) For any $x \in M, \alpha, \beta \in K$, we have $x(\alpha\beta) = (x\alpha)\beta$.
  (B2) For any $x \in M$, we have $x \cdot 1 = x$.

(C) These operations are compatible: multiplication is left and right distributive over addition:
  (C1) $x(\alpha + \beta) = x\alpha + x\beta$, for any $x \in M, \alpha, \beta \in K$.
  (C2) $(x + y)\alpha = x\alpha + y\alpha$, for any $x, y \in M, \alpha \in K$.

The product on M is called external because elements of M are multiplied by elements outside of M (lying in K) to obtain elements of M. Axiom (B1) is called ***mixed associativity*** because it involves two products, the one inside K and the external one on M. In (B2), 1 represents the identity of the ring K.

Left K***-modules*** are defined in the same way: in this case, the external multiplication, or left K-action, is a map $K \times M \to M, (\alpha, x) \mapsto \alpha x$, for $\alpha \in K, x \in M$. In order to emphasise the side on which external multiplication is defined, we denote a right K-module M as M_K, and a left K-module N as ${}_K N$.

Let M be a right K-module, with K not necessarily commutative. Then M has a left module structure over the opposite ring K^{op} : for any $x \in M, \alpha \in K$, set $\alpha \cdot x = x\alpha$, where the product on the right is calculated inside the right K-module M_K. It is easily verified that this action makes M a left K^{op}-module. So, right K-modules coincide with left K^{op}-modules. In the same way, left K-modules coincide with right K^{op}-modules. Because of that, it is equivalent to work with right or with left modules. If K is commutative, then $K = K^{\mathrm{op}}$ as rings, and therefore left and right K-modules coincide.

In order not to repeat statements, we have to fix a side, so we agree to work with right modules. From now on, and unless otherwise specified, the word 'module' means a right module.

I.2.13 Examples.

(a) If K is a field, then K-modules are just K-vector spaces. Indeed, the axioms for a module are the same as for a vector space. Thus, classical linear algebra is part of module theory.

(b) By definition, a module is an abelian group. Conversely, an abelian group M has a natural $\mathbb{Z}$-module structure. We must define an external operation $M \times \mathbb{Z} \to M$. For nonnegative integers, define it inductively: for $x \in M$, set $x0 = 0$ and, for $n \geq 1$, set $xn = x(n-1) + x$. If $n < 0$, set $xn = -(x\,|n|)$. It is easily seen that this multiplication satisfies conditions (B) and (C) of the definition and thus makes M a $\mathbb{Z}$-module. So, abelian groups coincide with $\mathbb{Z}$-modules (and abelian group theory is part of module theory).

(c) By definition, any module contains a zero element 0, so no module is empty. On the other hand, the one-element set $M = \{0\}$ is a K-module for the operations defined by $0+0 = 0$ and $0 \cdot \alpha = 0$, for any $\alpha \in K$. This is the ***zero*** or ***trivial*** module, denoted as $M = 0$.

(d) Any ring K is itself a right (and a left) K-module, the right (or left) K-action on K is the multiplication of the ring.

(e) The set $K[t]$ of polynomials in t with coefficients in a ring K is a right K-module under ordinary addition of polynomials and right multiplication of polynomials by elements of K. Also, $K[t]$ may be viewed as a left K-module. These two module structures coincide if K is commutative.

(f) For an integer $n \geq 1$, the set $M_n(K)$ of $n \times n$-matrices with coefficients in a ring K is a right K-module under ordinary addition of matrices and right multiplication of matrices by elements of K. Again, $M_n(K)$ may be considered as left K-module, and these two structures coincide when K is commutative.

We refrain from giving more examples because they arise more naturally later on. We need a notation. When dealing with a K-module M, we deal with two zeros, the one of M and the one of K. Usually, both are denoted by 0. When we want to distinguish them, we denote the first by 0_M and the second by 0_K. Arithmetic properties of modules resemble those of vector spaces, as is seen from the following statement.

I.2.14 Proposition. *Let K be a ring and M_K a module, then:*

(a) *The zero element of M is unique.*

(b) *The negative of any $x \in M$ is unique.*

(c) *For any $\alpha \in K$, we have $0_M \cdot \alpha = 0_M$.*

(d) *For any $x \in M$, we have $x \cdot 0_K = 0_M$.*

(e) *For any $\alpha \in K, x \in M$, we have $(-x)\alpha = x(-\alpha) = -(x\alpha)$.*

(f) *For any $\alpha \in K, x_1, \ldots, x_m \in M$, we have $(\sum_{i=1}^m x_i)\alpha = \sum_{i=1}^m (x_i\alpha)$.*

(g) *For any $\alpha_1, \ldots, \alpha_n \in K, x \in M$, we have $x(\sum_{j=1}^n \alpha_j) = \sum_{j=1}^n (x\alpha_j)$.*

(h) *For any $\alpha \in K, x \in M, n \in \mathbb{Z}$, we have $x(n\alpha) = (x\alpha)n = (xn)\alpha$.*

Proof. The proofs of statements (a) to (g) are similar to those of the corresponding statements in Proposition I.2.3, so they are left to the reader. We just prove (h). If $n = 0$, then equality follows from (c) and (d). If $n > 0$, it follows from (f) and (g). Assume that $n < 0$. Because of (e), we have:

$$(xn)\alpha = (-x\,|n|)\alpha = -x(|n|\,\alpha) = x(-\,|n|\,\alpha) = x(n\alpha).$$

Similarly, $(xn)\alpha = (x\alpha)n$. □

The abelian group structure of a module M implies that we also have $-(-x) = x$ and $-(x+y) = (-x) + (-y)$, for $x, y \in M$.

Together with a mathematical object come its subobjects.

I.2.15 Definition. Let K be a ring and M_K be a module. A ***submodule*** L of M is a subset of M which is itself a module under the same operations as M.

Thus, if L is a submodule of a module M and $x, y \in L$, then their sum in L is the same as their sum in M and, if $\alpha \in K$, the product of x by α in L is the same as their product in M. This is expressed by writing $x + y \in L$ and $x\alpha \in L$, respectively. A consequence is that, if $x, y \in L$, and $\alpha, \beta \in K$, then $x\alpha + y\beta \in L$. We prove that, conversely, this last condition on a nonempty subset L of a module M implies that L is a submodule of M.

I.2.16 Proposition. *Let M be a K-module and L a nonempty subset of M. Then L is a submodule of M if and only if for any $x, y \in L$, and $\alpha, \beta \in K$, we have $x\alpha + y\beta \in L$.*

Proof. We have just shown necessity so we prove sufficiency. The hypothesis (setting first $\alpha = \beta = 1$, then $\beta = 0$) guarantees the existence of operations in L. We check that the module axioms are satisfied in L. Identities (A1), (A2), (B1), (B2), (C1) and (C2) are satisfied in L because they are satisfied in M, so we just have to check (A3) and (A4). Because L is nonempty, there exists $x \in L$. The hypothesis gives $x \cdot 0 = 0 \in L$, hence L contains the zero of M (which is thus a zero element in L). Similarly, if $x \in L$, then $x(-1) = -x \in L$. The proof is complete. □

In a module M, an expression of the form $x\alpha + y\beta$ as above is called a ***linear combination*** of x, y with coefficients $\alpha, \beta \in K$. The proposition says that a nonempty subset is a submodule if and only if it is ***stable***, or ***closed***, under linear combinations.

Again, examples of submodules will appear more naturally later on. For the time being, we content ourselves with obvious ones.

I.2.17 Examples.

(a) Any module M has two submodules, itself and the set $\{0\}$. These coincide if and only if $M = 0$.

(b) If K is a field, then a submodule of a K-module (=vector space) is a subspace. Indeed, the condition of Proposition I.2.16 is the well-known criterion allowing to verify whether a subset of a vector space is a subspace or not.

(c) If $K = \mathbb{Z}$, that is, the module is an abelian group, then a submodule is a subgroup. Actually, one proves easily that the condition in Proposition I.2.16 reformulates as follows: a nonempty subset L of an abelian group M is a submodule if and only if, for any $x, y \in L$, we have $x - y \in L$.

Exercises of Section I.2

1. (a) Let K be a ring. By developing $(\alpha + \beta)(1 + 1)$ in two different ways, show that commutativity of addition follows from the other axioms for rings.
(b) State and prove the corresponding statement for K-modules.

2. Let K be a ring.
 (a) If $\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n$ are elements of K, prove that:

$$\left(\sum_{i=1}^{m} \alpha_i\right)\left(\sum_{j=1}^{n} \beta_j\right) = \sum_{i=1}^{m}\sum_{j=1}^{n}(\alpha_i\beta_j).$$

 (b) If $\alpha, \beta \in K$ satisfy $\alpha\beta = \beta\alpha$, prove that $\alpha^m\beta^n = \beta^n\alpha^m$, for any integers $m, n \geq 0$.
 (c) If $\alpha, \beta \in K$ satisfy $\alpha\beta = \beta\alpha$, prove that, for any integer $n > 0$, the ***binomial theorem*** holds true

$$(\alpha + \beta)^n = \sum_{i=0}^{n}\binom{n}{i}\alpha^i\beta^{n-i}.$$

3. A ring K in which $\alpha^2 = \alpha$, for any $\alpha \in K$, is called a ***boolean ring***.
 (a) Let K be a boolean ring. Prove that $\alpha = -\alpha$, for any $\alpha \in K$.
 (b) Prove that any boolean ring is commutative.
 (c) Let E be a set and $A = \mathcal{P}(E)$ its power set (that is, the set of its subsets). For $X, Y \in A$, define $X + Y = X \Delta Y$, where Δ denotes the symmetric difference between the two sets, and $XY = X \cap Y$. Prove that A, together with these operations, is a boolean ring.

4. Let K, K' be rings. We define on the cartesian product $K \times K' = \{(\alpha, \alpha') : \alpha \in K, \alpha' \in K'\}$ an addition and a multiplication componentwise, that is, we set

$$(\alpha, \alpha') + (\beta, \beta') = (\alpha + \beta, \alpha' + \beta') \quad \text{and} \quad (\alpha, \alpha') \cdot (\beta, \beta') = (\alpha\beta, \alpha'\beta')$$

for $\alpha, \beta \in K$ and $\alpha', \beta' \in K'$. Show that:
 (a) $K \times K'$, endowed with these operations, is a ring, called the ***product*** of K and K'.
 (b) $K \times K'$ is commutative if K and K' are commutative.
 (c) The product of two nonzero integral domains is not an integral domain.

5. Let K be an integral domain. If there exists a nonzero integer n such that $n\alpha = 0$ for any $\alpha \in K$, the least such positive n is called the ***characteristic*** of K and denoted by char$K$. If no such integer exists, we say that K has characteristic zero and write char$K = 0$. For example, char$\mathbb{R} = 0$.

(a) Prove that $n\alpha = 0$, for any $\alpha \in K$, if and only if $n \cdot 1 = 0$.
(b) Prove that the characteristic of an integral domain is either zero or a prime number.
(c) Let p be a prime integer. Prove that $\mathrm{char}(\mathbb{Z}/p\mathbb{Z}) = p$.
(d) Assume that char$K = p$ is a prime integer. Prove that, for any $\alpha, \beta \in K$, we have $(\alpha + \beta)^p = \alpha^p + \beta^p$. Deduce that, in this case, the map $\varphi : K \to K$ given by $\varphi(\alpha) = \alpha^p$, for $\alpha \in K$, is a ring homomorphism.
(e) Assume that char$K = p$ is a prime integer and let $\alpha \in K$ be such that there exists n, which is not a multiple of p, satisfying $n\alpha = 0$. Prove that $\alpha = 0$.

6. Prove that a ring in which any nonzero element has a right inverse is a division ring.

7. Using quaternions, prove that, if two integers m, n are sums of 4 squares (of integers), then so is their product mn.

8. Let $\mathbb{Z}[i] = \{\alpha + i\beta : \alpha, \beta \in \mathbb{Z}\}$ be the ring of ***Gaussian integers***. Prove that $\mathbb{Z}[i]$ is an integral domain and that its field of fractions is $\mathbb{Q}[i] = \{\lambda + i\mu : \lambda, \mu \in \mathbb{Q}\}$.

9. Prove that the set of real matrices of the form

$$\begin{pmatrix} a & b & c & d \\ -b & a & -d & c \\ -c & d & a & -b \\ -d & -c & b & a \end{pmatrix}$$

equipped with the usual addition and multiplication of matrices is a noncommutative division ring.

10. Let K be a commutative ring, $n > 0$ an integer, $M_n(K)$ the ring of $n \times n$ matrices with coefficients in K and K^n the abelian group $K^n = \{(\alpha_1, \ldots, \alpha_n) : \alpha_i \in K\}$, where addition is defined componentwise.

(a) Prove that ordinary matrix multiplication of an element of K^n by a matrix induces on K^n the structure of a right $M_n(K)$-module.
(b) By considering column instead of row vectors in K^n, define a left $M_n(K)$-module structure on K^n.

11. Let $n > 1$ be an integer and G an additive abelian group. Prove that G has a natural $\mathbb{Z}/n\mathbb{Z}$-module structure defined by $x\bar{a} = xa$, for $x \in G$ and $\bar{a} = a + n\mathbb{Z} \in \mathbb{Z}/n\mathbb{Z}$, if and only if $nx = 0$, for any $x \in G$.

12. Let G be an abelian group and $K = \mathrm{End}_{\mathbb{Z}} G$ the set of all its endomorphisms (that is, group homomorphisms from G to itself). Prove that:

(a) K is an abelian group under the addition defined by $(f+g)(x) = f(x)+g(x)$, for $f, g \in K, x \in G$.
(b) K is a monoid under the usual composition of maps. Deduce that K is a ring.
(c) G is a left K-module under the multiplication defined by $f \cdot x = f(x)$ for $f \in K, x \in G$.

13. Let K be a ring. Show that the set of invertible elements of K is a group under the multiplication of K. Is it true that this group is abelian if and only if K is commutative?

I.3 Algebras

I.3.1 The definition of an algebra

It is sometimes useful to look at a ring as having an enriched structure. It is then called algebra. Because algebras have a richer structure than rings, working with the former offers more technical possibilities than working with the latter.

From now on, unless otherwise specified, the letter K stands for a commutative ring. A K-algebra A is a K-module together with an operation $A \times A \to A$, called multiplication, which is right and left distributive over addition and compatible with the K-action on A, that is,

$$(ab)\alpha = a(b\alpha) = (a\alpha)b$$

for $a, b \in A$ and $\alpha \in K$. Because the ring K is commutative, the right K-module A is also a left K-module, that is, one can write the multiplication on A by K on either side. We simply say that A is a K-module. Listing the axioms for algebras yields the following definition.

I.3.1 Definition. Let K be a commutative ring. A set A is a ***K-algebra*** or ***algebra over K*** if:

(A) There is an operation $+$ on A which associates to each pair $(a, b) \in A \times A$ its ***sum*** $a + b \in A$, and which makes the pair $(A, +)$ an abelian group, that is:

(A1) It is commutative: $a + b = b + a$, for any $a, b \in A$.

(A2) It is associative: $a + (b + c) = (a + b) + c$, for any $a, b, c \in A$.

(A3) There exists in A an element called the ***zero***, and denoted by 0 or 0_A, such that $a + 0 = 0 + a = a$, for any $a \in A$.

(A4) To any element $a \in A$ is associated its ***negative***, or ***opposite***, $(-a)$ which is such that $(-a) + a = a + (-a) = 0$.

(B) The set A is equipped with an external operation, or right K-action, associating to each pair $(a, \alpha) \in A \times K$ an element $a\alpha \in A$, called its ***external product***, and making A a K-module, that is:

(B1) $a(\alpha\beta) = (a\alpha)\beta$, for any $a \in A, \alpha, \beta \in K$.

(B2) $a \cdot 1_K = a$, for any $a \in A$.

(C) External multiplication is left and right distributive over addition:

(C1) $(a + b)\alpha = a\alpha + b\alpha$, for any $a, b \in A, \alpha \in K$.

(C2) $a(\alpha + \beta) = a\alpha + a\beta$, for any $a \in A, \alpha, \beta \in K$.

(D) The set A is equipped with an ***internal multiplication*** associating to each pair $(a, b) \in A \times A$ an element $a \cdot b$, or simply ab, in A, called its ***product***, and which is compatible with the two previous operations:

(D1) $(a + b)c = ac + bc$, for any $a, b, c \in A$.
(D2) $c(a + b) = ca + cb$, for any $a, b, c \in A$.
(D3) $(ab)\alpha = a(b\alpha) = (a\alpha)b$, for any $a, b \in A, \alpha \in K$.

In axiom (B2), 1_K denotes the (multiplicative) identity of the ring K. We did not assume that the multiplication $A \times A \to A$ has an identity. If this is the case, that is, if there exists an element 1_A, or 1, in A such that $a \cdot 1_A = 1_A \cdot a = a$, for any $a \in A$, then A is called a ***unitary algebra***. Similarly, if the multiplication $A \times A \to A$ is associative, that is, if $a(bc) = (ab)c$, for any $a, b, c \in A$, then A is called an ***associative algebra***. In this text, unless otherwise specified, the word 'algebra' means an associative and unitary algebra.

The important restriction here is associativity. The existence of an identity is more an apparent constraint than a real one, because any algebra can be embedded into a unitary one, see Exercises I.3.1 and I.4.2, and modules over these two algebras can be considered to be the same, see Exercises II.2.1 and II.3.20. Thus, an algebra is at the same time a ring, in the sense of Definition I.2.1, and a K-module, in the sense of Definition I.2.12, these two structures being compatible.

An algebra A is called a ***commutative algebra*** if the multiplication $A \times A \to A$ is commutative, that is, if $ab = ba$, for any $a, b \in A$ (but we do not assume it in general).

Given a K-algebra A, one defines its ***opposite algebra*** A^{op} in the same way as the opposite ring: it is the same set, with the same K-module structure, but the product of the elements a, b is defined as $a \cdot b = ba$, where the product on the right is computed inside A. Thus, $A = A^{\mathrm{op}}$, *as algebras*, if and only if A is commutative.

We shall see in Subsection I.3.3 several examples of algebras, but here are three immediate ones. The zero K-module $A = \{0\}$ becomes an algebra if one defines multiplication by $0 \cdot 0 = 0$. This is the ***zero***, or ***trivial*** algebra, denoted (predictably) by $A = 0$. The set $K[t]$ of polynomials in one indeterminate t with coefficients in K has a K-module structure, see Example I.2.13(e), and also a ring structure, see Example I.2.2(d). Because these structures are compatible, that is, axiom (D3) is satisfied, $K[t]$ is an algebra, obviously commutative, called the ***algebra of polynomials*** in t over K. Given an integer $n > 0$, the set $M_n(K)$ of $n \times n$-matrices with coefficients in K has a K-module structure, see Example I.2.13(f), and also a ring structure, see Example I.2.2(e). Because these structures are compatible, $M_n(K)$ is an algebra, usually not commutative, called the ***full matrix algebra*** over K.

Because an algebra is at the same time a ring and a module, verifying the extra compatibility condition (D3), then it satisfies all arithmetic properties listed in Propositions I.2.3 and I.2.14.

As defined, algebras are particular types of rings. On the other hand, any ring can be viewed as an algebra over its centre $Z(A) = \{a \in A : xa = ax \text{ for any } x \in A\}$ or over the integers $\mathbb{Z}$. This follows from the next proposition which, given a commutative ring K, says that a ring is a K-algebra if and only if there exists a ring homomorphism from K to the centre of the ring.

I.3.2 Proposition. *Let K be a commutative ring.*

(a) *If A is a K-algebra, then the map $\varphi : K \to A$ defined by $\alpha \mapsto 1_A \cdot \alpha$, for $\alpha \in K$, is a ring homomorphism whose image is contained in the centre $Z(A)$ of A.*

(b) *Let A be a ring and $\varphi : K \to Z(A)$ a ring homomorphism. The external multiplication $A \times K \to A, (a, \alpha) \mapsto a\varphi(\alpha)$, for $a \in A, \alpha \in K$, gives A a K-algebra structure.*

Proof. (a) It is easily seen that φ is a ring homomorphism. We prove that its image is contained in $Z(A)$. Let $\alpha \in K$ and $a \in A$, then we have:

$$a\varphi(\alpha) = a(1_A \cdot \alpha) = (a\alpha)1_A = a\alpha = 1_A(a\alpha) = (1_A \cdot \alpha)a = \varphi(\alpha)a$$

where we used axiom (D3) at the second and fifth equality.

(b) It is easily verified that the given K-action makes A a K-module. We prove compatibility of this module structure with the ring structure of A. Let $a, b \in A$ and $\alpha \in K$, then:

$$a(b\alpha) = a(b\varphi(\alpha)) = a(\varphi(\alpha)b) = (a\varphi(\alpha))b = (a\alpha)b$$

because $\varphi(\alpha) \in Z(A)$. Also, we have:

$$(ab)\alpha = (ab)\varphi(\alpha) = a(b\varphi(\alpha)) = a(b\alpha)$$

because multiplication is associative in the ring A. □

Taking in (b), $K = Z(A)$ and $\varphi = 1_{Z(A)}$, we deduce that any ring is an algebra over its centre. Also, taking $K = \mathbb{Z}$, and $\varphi : n \mapsto 1_A \cdot n$, we get that any ring is a $\mathbb{Z}$-algebra. Thus, the two concepts of ring and algebra may be considered equivalent. This allows us to use the terminology of algebras.

In the notation of the proposition, $\varphi(0_K) = 0_A$ and $\varphi(1_K) = 1_A$. We may thus identify the zeros of K and A, as well as their identities, and denote them simply by 0 and 1, respectively. In fact, from now on, we use the same symbols 0 and 1 for all rings and algebras under consideration. This will entail no confusion.

If K is a field and A is nonzero, then the nonzero map $\varphi : K \to A$ is necessarily injective (because of Proposition I.3.9 below), and so K may be identified to a subfield of A, contained inside its centre. Also, A is a K-vector space, so we may talk about its dimension $\dim_K A$, defined, as usual, as the cardinality of an arbitrary basis in A. The algebra A is said to be a ***finite-dimensional algebra*** if $\dim_K A < \infty$, and an ***infinite-dimensional algebra*** otherwise. For instance, the set $\mathbb{C}$ of complex numbers is an algebra over the set $\mathbb{R}$ of the reals. It is two-dimensional, an obvious basis being $\{1, i\}$.

I.3.2 Subalgebras and ideals

It is natural, when one defines an algebraic structure to study the corresponding substructures, as we did for modules and submodules in Subsection I.2.2.

I.3.3 Definition. Let A be a K-algebra. A subset B of A is a ***subalgebra*** if it is an algebra for the same operations as A.

In particular, a subalgebra is never empty, because it contains the zero and the identity of A. While any algebra A is a subalgebra of itself, one sees that, if $A \neq 0$, then the subset $\{0\}$ is *not* a subalgebra, because it does not contain the identity of A.

The next criterion allows us to verify whether a subset of an algebra is a subalgebra or not.

I.3.4 Lemma. *Let A be an algebra. A subset B of A is a subalgebra if and only if the following conditions are satisfied:*

(a) $1 \in B$,

(b) *B is stable under K-linear combinations, that is, if $a, b \in B, \alpha, \beta \in K$, then $a\alpha + b\beta \in B$,*

(c) *B is stable under multiplication, that is, if $a, b \in B$, then $ab \in B$.*

Proof. Necessity is obvious. Sufficiency follows from the fact that the stated conditions guarantee that B is nonempty and the result of any operation in A performed between elements of B lies in B. Algebra axioms are satisfied in B because they are satisfied in A. □

Clearly, if A is commutative, then so is any of its subalgebras.

A consequence of the previous lemma is that the intersection of any family of subalgebras is also a subalgebra. In particular, the intersection of all subalgebras of A containing a given subset $X \subseteq A$ is itself a subalgebra of A, called the subalgebra ***generated*** by X.

In practice, the notion of ideal is more important than that of subalgebra.

I.3.5 Definition. Let A be a K-algebra. An ***ideal*** or, more precisely, ***two-sided ideal***, I of A is a K-submodule of A such that $x \in I, a \in A$ imply $xa \in I$ and $ax \in I$.

That is, ideals are additive subgroups which are right and left stable under multiplication by elements of the algebra. The fact that I is an ideal of A is expressed by the notation $I \trianglelefteq A$.

Considering a ring A as a $\mathbb{Z}$-algebra, we get the classical definition of ideal in the ring A. Conversely, if A is a K-algebra, the notion of ideal of A considered as a ring coincides with that of ideal of A as an algebra: indeed, let I be an ideal in A as a ring, in order to show that I is an ideal in A as an algebra, it suffices to show that I is a K-submodule of A, that is, it is stable under the right K-action. Now, for any $x \in I, \alpha \in K$, one has

$$x\alpha = (1 \cdot x)\alpha = (1 \cdot \alpha)x \in I$$

due to axiom (D3), as required. Any algebra has at least two ideals, namely itself and $\{0\}$. The only ideal containing the identity 1 of an algebra A is A itself: for, if $I \trianglelefteq A$ and $1 \in I$, then, for any $a \in A$, we have $a = a \cdot 1 \in I$ and so $I = A$.

Fundamental construction procedures for ideals starting from others are intersection and sum.

I.3.6 Lemma. *Let $(I_\lambda)_{\lambda\in\Lambda}$ be a family of ideals, then so is their intersection $\cap_\lambda I_\lambda$.*

Proof. Because $0 \in I_\lambda$, for any λ, we have $0 \in \cap_\lambda I_\lambda$, so $\cap_\lambda I_\lambda \neq \emptyset$. If $x, y \in \cap_\lambda I_\lambda$, and $\alpha, \beta \in K$, we have $x\alpha + y\beta \in I_\lambda$, for any λ, because each I_λ is a K-submodule of A. Therefore, $x\alpha + y\beta \in \cap_\lambda I_\lambda$. On the other hand, if $a \in A$, then $ax, xa \in I_\lambda$, for any λ, because each I_λ is an ideal of A. Therefore, $ax, xa \in \cap_\lambda I_\lambda$, and the proof is complete. □

Consequently, the intersection of all ideals containing a given subset X of A is itself an ideal containing X. It is the smallest ideal containing X and for this reason is said to be ***generated*** by X. For instance, the ideal generated by the empty set is the zero ideal $\{0\}$. Given a nonempty set, one can describe explicitly the ideal it generates.

I.3.7 Lemma. *Let $X \subseteq A$ be nonempty. The ideal it generates is the set*

$$\langle X\rangle = \left\{\sum_{i=1}^{n} a_i x_i b_i : n \geq 0, x_i \in X, a_i, b_i \in A\right\}.$$

Proof. We leave to the reader the verification that the given set $\langle X\rangle$ is an ideal. We prove it is the smallest one containing X. If $x \in X$, then $x = 1 \cdot x \cdot 1$ shows that $X \subseteq \langle X\rangle$. Let I be an ideal containing X, and $y \in \langle X\rangle$. Then there exist an integer $n \geq 0, x_i \in X$ and elements $a_i, b_i \in A$ such that $y = \sum_{i=1}^{n} a_i x_i b_i$. Because each x_i lies in I, which is an ideal, we have $a_i x_i b_i \in I$, for any i. Hence, $\sum_{i=1}^{n} a_i x_i b_i \in I$ and $\langle X\rangle \subseteq I$, as required. □

If A is commutative, then the result of the lemma takes on a particularly nice form: in this case, $\langle X\rangle$ is the set of linear combinations $\sum_{i=1}^{n} c_i x_i$, with $n \geq 0, x_i \in X$ and $c_i \in A$. If, in particular, $X = \{x\}$ is a one-element set, then $\langle X\rangle = \langle x\rangle$ is the set $Ax = \{cx : c \in A\}$ of the multiples of x. Such an ideal Ax, generated by a single element, is called a ***principal ideal***. For instance, it is well-known that any subgroup of $\mathbb{Z}$ is of the form $n\mathbb{Z}$, for some integer n. Because each $n\mathbb{Z}$ is an ideal when $\mathbb{Z}$ is considered as an algebra, any ideal of $\mathbb{Z}$ is of the form $n\mathbb{Z}$, that is, is principal.

The sum of ideals is defined in the same way as the sum of subspaces of a vector space (or of subgroups of an abelian group). The ***support*** of a family of elements $(x_\lambda)_{\lambda\in\Lambda}$ in A is the set $\{\lambda \in \Lambda : x_\lambda \neq 0\}$. A family $(x_\lambda)_{\lambda\in\Lambda}$ of elements of A is called ***of finite support*** if its support is a finite set, or, equivalently, if $x_\lambda = 0$ for all but at most finitely many $\lambda \in \Lambda$. We use the suggestive expression $x_\lambda = 0$ ***for almost all*** λ. If $(x_\lambda)_{\lambda\in\Lambda}$ is such that $x_\lambda = 0$ for almost all λ, then the seemingly infinite sum $\sum_{\lambda\in\Lambda} x_\lambda$ is actually finite and makes sense in the algebra.

I.3.8 Definition. Let $(I_\lambda)_{\lambda\in\Lambda}$ be a family of ideals of an algebra A. Their ***sum*** is the set of the finite sums of the $x_\lambda \in I_\lambda$, that is:

$$\sum_{\lambda\in\Lambda} I_\lambda = \left\{\sum_{\lambda} x_\lambda : x_\lambda \in I_\lambda, \text{ for any } \lambda, \text{ and } x_\lambda = 0, \text{ for almost all } \lambda\right\}.$$

We leave to the reader the straightforward verification that $\sum_\lambda I_\lambda$ is an ideal whenever each I_λ is so, and, actually, is the smallest one containing all the I_λ.

We need another construction. Let I, J be ideals of an algebra A, then the set IJ of all sums of the form $\sum_\lambda x_\lambda y_\lambda$, where $x_\lambda \in I, y_\lambda \in J$ are such that $x_\lambda y_\lambda = 0$ for almost all λ, that is, of all finite sums of the form $\sum_\lambda x_\lambda y_\lambda$, is an ideal, called the ***product*** of I and J. Clearly, $IJ \subseteq I \cap J$. But in general, equality does not hold true, see Exercise I.3.10(d).

This allows us to define inductively the ***powers*** of an ideal I: we set $I^1 = I$ and $I^n = I^{n-1}I$ for any integer $n > 1$. Each of the I^n is an ideal.

One can characterise fields in terms of their ideals.

I.3.9 Proposition. *A nonzero commutative algebra is a field if and only if it has no nonzero proper ideals.*

Proof. Necessity. Let A be a field and I a nonzero ideal of A. Let $a \in I, a \neq 0$. Because A is a field, a^{-1} exists in A. Therefore, $1 = a^{-1}a \in I$, because I is an ideal. But then $I = A$.

Sufficiency. Assume that A has no nonzero proper ideal. Let $a \in A$ be nonzero. The ideal Aa generated by a is nonzero. Hence, $Aa = A$. Therefore, $1 \in Aa$ and so a is invertible. □

The assumption about commutativity of A is essential in the statement: see Example I.3.11(b) below.

Ideals allow to define quotients of algebras, just as normal subgroups allow to define quotients of groups. Let A be an algebra and I an ideal of A. In particular, I is an abelian subgroup of A, so the relation defined by $a \equiv b \pmod I$ if and only if $a - b \in I$ is an equivalence relation on A. The equivalence class of $a \in A$ is the so-called ***coset*** $a + I = \{a + x : x \in I\}$. The element a is called a ***representative*** of the coset $a + I$. Coset representatives are usually not unique: indeed, one has $a \equiv b \pmod I$ if and only if $a + I = b + I$. The quotient set $A/I = \{a + I : a \in A\}$ is an abelian group under the operation (unambiguously) defined, for $a, b \in A$, by

$$(a + I) + (b + I) = (a + b) + I,$$

which can be expressed by saying that the sum of two cosets is the coset of the sum of their representatives. The zero element of the group A/I is the coset $0 + I = I$.

We prove that A/I has actually a K-algebra structure.

(a) We define the multiplication of $a + I \in A/I$ with $\alpha \in K$ by

$$(a + I)\alpha = a\alpha + I,$$

that is, the product of a coset by a scalar is the coset of the product of the representative by this scalar. This multiplication is unambiguously defined: $a + I = a' + I$ implies $a - a' \in I$, hence $a\alpha - a'\alpha = (a - a')\alpha \in I$, because I is a K-submodule of A. Therefore, $a\alpha + I = a'\alpha + I$.

(b) We define the multiplication of $a + I, b + I \in A/I$ by

$$(a + I)(b + I) = ab + I,$$

that is, the product of cosets is the coset of the product of their representatives. This operation is again unambiguously defined: $a + I = a' + I$ and $b + I = b' + I$ imply

$a - a', b - b' \in I$, hence $ab - a'b' = a(b - b') + (a - a')b \in I$, because I is an ideal. Therefore, $ab + I = a'b' + I$.

I.3.10 Proposition. *Let A be an algebra and I an ideal of A. Then A/I is an algebra under the operations defined, for $a, b \in A$ and $\alpha \in K$, by*

$$(a + I) + (b + I) = (a + b) + I,$$
$$(a + I)\alpha = a\alpha + I,$$
$$(a + I)(b + I) = ab + I.$$

Proof. The proof consists in verifying the axioms one by one. We prove, for example, associativity of (internal) multiplication: let $a, b, c \in A$ then $a(bc) = (ab)c$ in A implies that:

$$\begin{aligned}(a + I)((b + I)(c + I)) &= (a + I)(bc + I) = a(bc) + I = (ab)c + I \\ &= (ab + I)(c + I) = ((a + I)(b + I))(c + I)\end{aligned}$$

as required. The class $1 + I$ is the identity of A/I, because, for any $a \in A$,

$$(a + I)(1 + I) = (a \cdot 1) + I = a + I = (1 \cdot a) + I = (1 + I)(a + I). \qquad \square$$

The algebra A/I is called the ***quotient algebra*** of A by I. Its zero element is the class $0 + I = I$.

The proof of the proposition illustrates a useful fact: A/I inherits many properties of A. For example, if A is commutative, then so is A/I, because, if $ab = ba$ in A, then $(a + I)(b + I) = ab + I = ba + I = (b + I)(a + I)$. Again, ‘many’ does not mean all. For instance, quotients of integral domains do not have to be integral: $\mathbb{Z}$ is an integral domain, but not $\mathbb{Z}/6\mathbb{Z}$.

The quotient may have additional desirable properties that A does not possess. For instance, the fact that I is the zero in A/I implies that a class $a + I$ equals the zero class if and only if $a \in I$: that is, elements of I are equated to zero in the quotient.

I.3.3 Examples of algebras

I.3.11 Examples.

(a) Consider the ring $\mathbb{H}$ of Hamilton’s quaternions, see Example I.2.8(a). Any quaternion can be written uniquely in the form

$$\begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix}$$

with $a, b, c, d \in \mathbb{R}$. Therefore, $\mathbb{H}$ is a four-dimensional real vector space. Because it is a ring, and matrix multiplication is compatible with multiplication by real numbers, it is a four-dimensional real algebra. An algebra which is also a division ring (such as $\mathbb{H}$) is called a ***division algebra***.

(b) Let A be a K-algebra, $n > 0$ an integer and $M_n(A)$ the set of $n \times n$-matrices with coefficients in A. Under ordinary matrix operations, $M_n(A)$ is a K-algebra. A particular role is played by the matrices $\mathbf{e}_{ij}$ having coefficient 1 at the intersection of line i and column j, and 0 elsewhere. Indeed, if $\mathbf{a} = [a_{ij}] \in M_n(A)$, then $\mathbf{a} = \sum_{i,j=1}^{n} \mathbf{e}_{ij} a_{ij}$, that is, any matrix can be written (uniquely!) as a linear combination of the $\mathbf{e}_{ij}$.

The matrices $\mathbf{e}_{ij}$ multiply as follows:

$$\mathbf{e}_{ij}\mathbf{e}_{kl} = \begin{cases} \mathbf{e}_{il} & \text{if } j = k, \\ 0 & \text{otherwise.} \end{cases}$$

If $A = K$ is a field (or, more generally, a division ring), then the matrices $\mathbf{e}_{ij}$ form a basis of the K-vector space $M_n(K)$. An important property of $M_n(K)$ is that this (noncommutative) algebra has no proper nonzero ideal (hence, Proposition I.3.9 does not extend to the noncommutative case). Let indeed $I \trianglelefteq M_n(K)$ be nonzero. Then there exists a nonzero matrix $\mathbf{a} = [a_{ij}]$ in I. Let r, s be such that $a_{rs} \neq 0$. Then a_{rs}^{-1} exists in K. But then, for any pair of indices i, j, we have:

$$\mathbf{e}_{ij} = a_{rs}^{-1}(\mathbf{e}_{ir}\mathbf{a}\mathbf{e}_{sj}) \in I$$

because I is an ideal. All basis vectors of $M_n(K)$ thus lie in I, hence $I = M_n(K)$.

(c) Several subalgebras of $M_n(K)$ are of interest. For instance, let $T_n(K) = \{\mathbf{a} = [a_{ij}] \in M_n(K) : a_{ij} = 0 \text{ for } i < j\}$ be the set of lower triangular matrices. Applying Lemma I.3.4, one verifies easily that it is a subalgebra of $M_n(K)$. Another example, for $n = 4$, is the set of matrices

$$A = \begin{pmatrix} K & 0 & 0 & 0 \\ K & K & 0 & 0 \\ K & 0 & K & 0 \\ K & K & K & K \end{pmatrix} = \left\{ \begin{pmatrix} \alpha_1 & 0 & 0 & 0 \\ \alpha_2 & \alpha_3 & 0 & 0 \\ \alpha_4 & 0 & \alpha_5 & 0 \\ \alpha_6 & \alpha_7 & \alpha_8 & \alpha_9 \end{pmatrix} : \alpha_i \in K, \text{ for all } i \right\}$$

which is a subalgebra of $T_4(K)$, and hence of $M_4(K)$.

(d) Let $A_1, \cdots, A_n$ be K-algebras. Their cartesian product

$$\prod_{i=1}^{n} A_i = A_1 \times \cdots \times A_n = \{(a_1, \cdots, a_n) : a_i \in A_i\}$$

admits a K-algebra structure, with operations defined componentwise

$$(a_1, \cdots, a_n) + (b_1, \cdots, b_n) = (a_1 + b_1, \cdots, a_n + b_n)$$
$$(a_1, \cdots, a_n)\alpha = (a_1\alpha, \cdots, a_n\alpha)$$
$$(a_1, \cdots, a_n)(b_1, \cdots, b_n) = (a_1b_1, \cdots, a_nb_n)$$

where $a_i, b_i \in A_i$, for each i, and $\alpha \in K$. The algebra $\prod_{i=1}^{n} A_i$ is the ***product*** of the A_i. If $A_1 = \cdots = A_n = A$, then we denote the product by A^n.

(e) One also considers infinite products of algebras. Let K be a commutative ring. A ***sequence*** of elements of K is a map $a : \mathbb{N} \to K$. We write $a(i) = a_i$, for $i \in \mathbb{N}$, and denote the sequence as $a = (a_i)_{i \geq 0}$. The set $K^{\mathbb{N}}$ of sequences of elements of K becomes a K-algebra if we set

$$(a_i)_{i \geq 0} + (b_i)_{i \geq 0} = (a_i + b_i)_{i \geq 0}$$
$$(a_i)_{i \geq 0}\alpha = (a_i\alpha)_{i \geq 0}$$
$$(a_i)_{i \geq 0}(b_i)_{i \geq 0} = \left(\sum_i a_j b_{i-j}\right)_{i \geq 0}$$

with $a_i, b_i \in K$, for any i, and $\alpha \in K$. The rule for multiplication being similar to that for polynomials, we agree to denote the previous sequence as $(a_i)_{i \geq 0} = \sum_{i=0}^{\infty} a_i t^i$ with t an indeterminate. The notation of sum clearly makes no sense here, because one does not assume the a_i almost all zero, but it makes calculations easier: in order to multiply two such elements, we multiply term-by-term, then we group together the summands corresponding to the same powers of t. The resulting algebra is denoted by $K[[t]]$ and called the ***algebra of formal power series*** in an indeterminate t, with coefficients in K. This algebra is commutative (because so is K) and contains the polynomial algebra $K[t]$ as a subalgebra.

(f) Let E be a finite poset (that is, partially ordered set), ordered by $\leqslant$. The ***incidence algebra*** KE of E is the set of linear combinations of the pairs $e_{ij} = (i, j) \in E \times E$, with $j \leqslant i$, having coefficients in K. The product of the pairs e_{ij} and e_{kl} is defined to be $e_{ij}e_{kl} = e_{il}$ if $j = k$ and zero otherwise. We then extend this product to arbitrary elements by distributivity. Thus, the e_{ij} multiply exactly as the matrices $\mathbf{e}_{ij}$ of Example (b) above. Consequently, KE can be viewed as a subalgebra of $M_n(K)$, where n is the cardinality of E. Actually, KE is a subalgebra of $T_n(K)$, because $i \leqslant j$ and $i \neq j$ imply $j \nleq i$.

For instance, if E is the finite poset $1 \leqslant 2 \leqslant \cdots \leqslant n$, then KE is another realisation of the lower triangular matrix algebra $T_n(K)$. Similarly, if $E = \{1, 2, 3, 4\}$, ordered by $1 \leqslant 2 \leqslant 4$, $1 \leqslant 3 \leqslant 4$ (and $2, 3$ non comparable), then KE is another realisation of the subalgebra A of $T_4(K)$ of Example (c) above.

(g) Let G be a group. The ***group algebra*** KG of G is the set of linear combinations of elements of G with coefficients in K. The product of $g \in G$ by $h \in G$ in KG is their product gh in the group G, and is extended to other elements of KG by distributivity. Thus, if $\sum_{g \in G} g\alpha_g$, $\sum_{h \in G} h\beta_h$ are elements of KG, where the elements α_g, β_h of K are almost all zero, then we have:

$$\left(\sum_{g \in G} g\alpha_g\right) \cdot \left(\sum_{h \in G} h\beta_h\right) = \sum_{k = gh} k\alpha_g\beta_h.$$

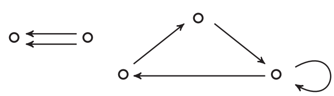
In Examples (e) and (f), we defined the algebra as the set of K-linear combinations of elements in a given generating set, we next defined multiplication on elements of this generating set, then we extended this definition by distributivity. The same technique is used in the next example.

(h) A ***quiver*** $Q = (Q_0, Q_1, s, t)$ is a quadruple consisting of two sets: Q_0, whose elements are called ***points***, and Q_1, whose elements are called ***arrows***, and two maps $s, t : Q_1 \to Q_0$ associating to an arrow $\alpha \in Q_1$ its ***source*** $s(\alpha)$ and its ***target*** $t(\alpha)$, both lying in Q_0.

An arrow α of source x and target y is denoted as $\alpha : x \to y$ or $x \xrightarrow{\alpha} y$. The quiver Q is called a ***finite quiver*** if Q_0, Q_1 are both finite sets. For an integer $\ell \geq 1$, a ***path*** $\alpha_1\alpha_2 \cdots \alpha_\ell$ of length ℓ from x to y is a sequence of ℓ arrows such that $s(\alpha_1) = x, t(\alpha_\ell) = y$ and $t(\alpha_i) = s(\alpha_{i+1})$, for all i satisfying $1 \leqslant i < \ell$. It may be depicted as

$$x = x_0 \xrightarrow{\alpha_1} x_1 \xrightarrow{\alpha_2} \cdots \xrightarrow{\alpha_\ell} x_\ell = y$$

(so, arrows are composed from left to right). We associate to each point $x \in Q_0$ a path ϵ_x of length zero from x to itself, called the ***stationary path*** at x. A quiver Q is called ***acyclic*** if it contains no cycle, that is, no path of length at least one from a point to itself. Examples of quivers are:



The first one is acyclic but not the second.

A ***subquiver*** $Q' = (Q'_0, Q'_1, s', t')$ of $Q = (Q_0, Q_1, s, t)$ is a quiver such that $Q'_0 \subseteq Q_0, Q'_1 \subseteq Q_1$ and if an arrow $x \xrightarrow{\alpha} y$ belongs to Q'_1, then $s'(\alpha) = x = s(\alpha), t'(\alpha) = y = t(\alpha)$, that is, the restrictions of s, t to Q'_1 equal s', t' respectively. We say that Q' is a ***full subquiver*** of Q provided $Q'_1 = \{\alpha \in Q_1 : s(\alpha), t(\alpha) \in Q'_0\}$, that is, the arrows of Q' are precisely those arrows of Q whose source and target belong to Q'. Thus, a full subquiver is completely determined by its set of points.

Let K be a commutative ring. To a finite quiver Q, we associate its ***path algebra*** KQ. The elements of KQ are the linear combinations of paths in Q, including the stationary ones. The product of paths is defined to be their composite if they are composable, and zero if they are not. That is, if $\alpha_1\alpha_2 \cdots \alpha_\ell$ and $\beta_1\beta_2 \cdots \beta_k$ are paths, then their product is

$$(\alpha_1\alpha_2 \cdots \alpha_\ell) \cdot (\beta_1\beta_2 \cdots \beta_k) = \begin{cases} \alpha_1\alpha_2 \cdots \alpha_\ell\beta_1\beta_2 \cdots \beta_k & \text{if } t(\alpha_\ell) = s(\beta_1), \\ 0 & \text{if } t(\alpha_\ell) \neq s(\beta_1). \end{cases}$$

It is then extended to all elements of KQ using distributivity. The identity is the sum $\sum_{x \in Q_0} \epsilon_x$ of all stationary paths, which exists in KQ because Q_0 is finite.

For instance, let K be a field and Q the quiver $3 \xrightarrow{\alpha} 2 \xrightarrow{\beta} 1$. A K-basis of KQ is given by the paths $\{\epsilon_1, \epsilon_2, \epsilon_3, \alpha, \beta, \alpha\beta\}$ with multiplication table given by:

	ϵ_1	ϵ_2	ϵ_3	α	β	$\alpha\beta$
ϵ_1	ϵ_1	0	0	0	0	0
ϵ_2	0	ϵ_2	0	0	β	0
ϵ_3	0	0	ϵ_3	α	0	$\alpha\beta$
α	0	α	0	0	$\alpha\beta$	0
β	β	0	0	0	0	0
$\alpha\beta$	$\alpha\beta$	0	0	0	0	0

In particular, $\dim_K KQ = 6$. Also, if Q is the quiver

$$\begin{tikzcd} \overset{\circ}{x} \arrow[loop above, "\alpha"] \end{tikzcd}$$

then KQ is generated by its unique stationary path ϵ_x, which is therefore the identity of KQ, and by the cycles α^i, for all $i \geq 1$. Multiplication of cycles is defined by the exponent rule $\alpha^i\alpha^j = \alpha^{i+j}$. In other words, KQ is another realisation of the polynomial algebra $K[t]$. In particular, if K is a field, then KQ is infinite-dimensional.

Because KQ is generated by its paths, it has finitely many generators if and only if Q is acyclic. If Q contains cycles, but we decide to equate to zero all paths long enough, then we still have only finitely many generators. This is achieved through factoring by an ideal containing these paths.

Let K be a field and KQ^+ the ideal of KQ generated by the arrows. It contains all paths of positive length. For an integer $m \geq 2$, let KQ^{+m} be the m^{th} power of KQ^+, see the definition before Proposition I.3.9. That is, KQ^{+m} is the ideal generated by paths of length m. It contains all paths of length greater than or equal to m. An ideal I of KQ is called ***admissible*** if there exists $m \geq 2$ such that

$$KQ^{+m} \subseteq I \subseteq KQ^{+2}.$$

To say that $I \subseteq KQ^{+2}$ means that I contains only linear combinations of paths of length at least two. Note that we suppose that $I \subseteq KQ^{+2}$ and not $I \subseteq KQ^+$: for, if an element of I belongs to $KQ^+ \setminus KQ^{+2}$, then at least one arrow of the quiver would be superfluous as a generator of KQ. To say that there exists $m \geq 2$ such that $KQ^{+m} \subseteq I$ means that all paths of length at least m (that is, long enough) belong to I.

If I is admissible, then the quotient algebra KQ/I is generated by the residual classes modulo I of the paths in Q, and only finitely many of them are nonzero in the quotient because $KQ^{+m} \subseteq I$. We call KQ/I a ***bound quiver algebra***. The pair (Q, I) is called a ***bound quiver***. We have just proved that bound quiver algebras have finite bases, that is, they are finite-dimensional.

For example, for the quiver $3 \xrightarrow{\alpha} 2 \xrightarrow{\beta} 1$ considered above, the ideal I generated by $\alpha\beta$ is admissible. The quotient KQ/I, in this case, has dimension 5, and a basis is given by the set $\{\epsilon_1 + I, \epsilon_2 + I, \epsilon_3 + I, \alpha + I, \beta + I\}$. In the one-loop quiver above, we can consider I to be the ideal $\langle \alpha^2 \rangle$ generated by α^2. So, I contains all α^i with $i \geq 2$, hence $(\alpha + I)^i = 0$ for $i \geq 2$, and the two elements $\epsilon_x + I, \alpha + I$ constitute a basis of KQ/I.

In a bound quiver algebra, one can visualise on the quiver a basis and also how to multiply its vectors. We shall see later that one can also express in terms of points and arrows many structural features of this class of algebras. This makes it particularly suitable for constructing illustrative examples.

For instance, it is easy to visualise the opposite algebra of a path algebra: given a quiver Q, construct the opposite quiver by keeping the same points and reversing the sense of arrows, then the path algebra of the opposite quiver is another realisation of the opposite algebra of the path algebra of Q, see Exercise I.4.8.

Exercises of Section I.3

1. Let A be an associative K-algebra which does not necessarily have a multiplicative identity. Consider the set $A_1 = A \times K = \{(a, \alpha) : a \in A, \alpha \in K\}$ equipped with the operations

$$\begin{aligned} (a,\alpha) + (b,\beta) &= (a+b, \alpha+\beta) \\ (a,\alpha)\beta &= (a\beta, \alpha\beta) \\ (a,\alpha)(b,\beta) &= (ab + a\beta + b\alpha, \alpha\beta) \end{aligned}$$

for $a, b \in A, \alpha, \beta \in K$. Verify that A_1 is an associative K-algebra having $(0, 1)$ as an identity (so, any associative algebra can be embedded in a unitary associative algebra). One says that A_1 is obtained from A by ***adjoining an identity***.

2. Let K be a field. Prove that each of the following sets of matrices, under the usual matrix operations, is a subalgebra of $T_4(K)$.

(a) $A_0 = \begin{pmatrix} K & 0 & 0 & 0 \\ 0 & K & 0 & 0 \\ 0 & 0 & K & 0 \\ 0 & 0 & 0 & K \end{pmatrix} = \left\{ \begin{pmatrix} \alpha & 0 & 0 & 0 \\ 0 & \beta & 0 & 0 \\ 0 & 0 & \gamma & 0 \\ 0 & 0 & 0 & \delta \end{pmatrix} : \alpha, \beta, \gamma, \delta \in K \right\}$

(b) $A_1 = \begin{pmatrix} K & 0 & 0 & 0 \\ 0 & K & 0 & 0 \\ 0 & 0 & K & 0 \\ K & K & K & K \end{pmatrix} = \left\{ \begin{pmatrix} \alpha_1 & 0 & 0 & 0 \\ 0 & \alpha_2 & 0 & 0 \\ 0 & 0 & \alpha_3 & 0 \\ \alpha_4 & \alpha_5 & \alpha_6 & \alpha_7 \end{pmatrix} : \alpha_i \in K \text{ for all } i \right\}$

(c) $A_2 = \left\{ \begin{pmatrix} \alpha & 0 & 0 & 0 \\ \beta & \alpha & 0 & 0 \\ \gamma & \beta & \alpha & 0 \\ \delta & \gamma & \beta & \alpha \end{pmatrix} : \alpha, \beta, \gamma, \delta \in K \right\}$

(d) $A_3 = \left\{ \begin{pmatrix} \alpha & 0 & 0 & 0 \\ 0 & \alpha & 0 & 0 \\ \beta & \gamma & \alpha & 0 \\ \delta & \epsilon & \lambda & \alpha \end{pmatrix} : \alpha, \beta, \gamma, \delta, \epsilon, \lambda \in K \right\}$

3. Let A be a K-algebra. Prove that the centre $Z(A)$ of A is a subalgebra of A.

4. We show an infinite-dimensional analogue of the lower triangular matrix algebra $T_n(K)$. A matrix $\mathbf{a} = (a_{ij})_{i,j\geq 1}$, with rows and columns indexed by the positive integers, is ***row-finite*** (or ***column-finite***) if it has at most finitely many nonzero coefficients in each row (or column, respectively). It is ***lower triangular*** if $a_{ij} = 0$ whenever $i < j$. Let $TF(K)$ be the set of lower triangular matrices with coefficients in K which are both row-finite and column-finite. Prove that:

(a) $TF(K)$ is a K-algebra under the usual matrix operations.

(b) The set of matrices $\mathbf{a} = (a_{ij})_{i,j\geq 1} \in TF(K)$ such that $a_{ii} = 0$, for any i, is an ideal in $TF(K)$.

5. Let B be at the same time a subalgebra and an ideal of an algebra A. Prove that $B = A$.

6. Prove that a nonempty subset I of a K-algebra A is an ideal if and only if, for any $x, y \in I, a, b, c, d \in A$, we have $axb + cyd \in I$.

7. Let I, J be ideals of an algebra A. Prove that $I \cup J$ is an ideal of A if and only if $I \subseteq J$ or $J \subseteq I$.

8. Let A be an algebra and $(I_\lambda)_{\lambda\in\Lambda}$ a family of ideals of A which is a ***chain***, that is, for any $\lambda, \mu \in \Lambda$, we have $I_\lambda \subseteq I_\mu$ or $I_\mu \subseteq I_\lambda$. Prove that $\cup_{\lambda\in\Lambda} I_\lambda$ is an ideal of A.

9. Let A, B be algebras. Prove that any ideal of the product $A \times B$ is of the form $I \times J$, for some $I \trianglelefteq A, J \trianglelefteq B$.

10. Let I, J, L be ideals of an algebra A. Prove that:

(a) $(IJ)L = I(JL)$.

(b) $I(J + L) = IJ + IL$ and $(J + L)I = JI + LI$.

(c) If $J \subseteq I$, then $I \cap (J + L) = J + (I \cap L)$. This identity is known as the ***modular law***.

(d) In general, $IJ \neq I \cap J$.

11. Let A be a commutative algebra and $X, Y \subseteq A$ subsets such that each element of one of them is a linear combination (with coefficients in A) of elements of the other. Prove that X and Y generate the same ideal in A.

12. Let A be an algebra and $I \trianglelefteq A$. Prove that A/I is commutative if and only if, for any $a, b \in A$, we have $ab - ba \in I$. Deduce that any algebra has a commutative quotient.

13. Let A be a commutative algebra. An element $a \in A$ is called ***nilpotent*** if there exists an integer $n > 0$ such that $a^n = 0$. Prove that the set N of nilpotent elements of A is an ideal of A and the quotient A/N contains no nonzero nilpotent elements.

14. Let K be a nonzero commutative ring. Prove that:
(a) There exists a unique ring homomorphism $\varphi : \mathbb{Z} \to K$.
(b) For $x \in \mathbb{Z}$, we have $\varphi(x) = 0$ if and only if x is divisible by the characteristic of K, if the latter is positive, and $x = 0$ otherwise. See Exercise I.2.5.

15. Let K be a commutative ring and $K[[t]]$ the algebra of formal power series in t, see Example I.3.11(e). The ***order*** $\omega(a)$ of a nonzero element $a = (a_i)_{i\geq 0} \in K[[t]]$ is the least nonnegative integer i_0 such that $a_{i_0} \neq 0$. Prove that:
(a) If $a, b \in K[[t]]$, then either $ab = 0$, or $\omega(ab) \geq \omega(a) + \omega(b)$, with equality if K is an integral domain.
(b) If $a, b \in K[[t]]$, then either $a + b = 0$, or $\omega(a + b) \geq \min\{\omega(a), \omega(b)\}$.
(c) If K in an integral domain, then so is $K[[t]]$.
(d) The element $1 - t$ is invertible in $K[[t]]$, and its inverse is $\sum_{i=0}^{\infty} t^i$.
(e) An element $a = (a_i)_{i\geq 0} \in K[[t]]$ is invertible in $K[[t]]$ if and only if a_0 is invertible in K.
(f) If K is a field, then any nonzero element of $K[[t]]$ can be written in the form $t^i a$, for some $i \geq 0$ and an invertible $a \in K[[t]]$.

16. Let K be a field and Q the quiver

$$\begin{tikzcd}
 & & \circ \arrow[dl, "\beta"'] & \\
\circ & \circ \arrow[l, "\alpha"'] & & \circ \arrow[ul, "\gamma"'] \arrow[dl, "\epsilon"] \\
 & & \circ \arrow[ul, "\delta"] &
\end{tikzcd}$$

(a) Prove that the path algebra KQ of Q is finite-dimensional and find its dimension.
(b) Show that the ideal $I = \langle \gamma\beta - \epsilon\delta \rangle$ of KQ is admissible and write a basis for $A = KQ/I$.
(c) Give generators for admissible ideals I_i, with $i = 1, 2$, such that the corresponding quotient algebras $A_i = KQ/I_i$ have dimensions 10 and 13, respectively.

I.4 Algebra morphisms

I.4.1 Morphisms

A homomorphism between sets having the same algebraic structure is a map which preserves the structure, in the sense that it is compatible with the operations defining it. An (associative and unitary) algebra is defined by three operations: an addition, an external multiplication and an internal one, the latter admitting an identity. We need compatibility with respect to each.

I.4.1 Definition. Let A, B be K-algebras. A ***morphism***, or ***homomorphism***, of K-algebras from A to B is a map $\varphi : A \to B$ satisfying the following conditions:

(AM1) $\varphi(a_1 + a_2) = \varphi(a_1) + \varphi(a_2)$,
(AM2) $\varphi(a\alpha) = \varphi(a)\alpha$,
(AM3) $\varphi(a_1 a_2) = \varphi(a_1)\varphi(a_2)$,
(AM4) $\varphi(1) = 1$,

for any $a, a_1, a_2 \in A, \alpha \in K$.

We use the term morphism, rather than homomorphism, not only for brevity but because it fits into a general scheme which we introduce in Chapter III. The reader certainly noticed that in (AM4), we used the same symbol 1 to denote the identities of A and B.

The previous definition says that a map is an algebra morphism if and only if it is a ring homomorphism satisfying the additional axiom (AM2), expressed by saying that it is K-linear.

If $K = \mathbb{Z}$, then axiom (AM2) is a consequence of (AM1) and induction. Because rings are $\mathbb{Z}$-algebras, this implies that any ring homomorphism is a morphism of $\mathbb{Z}$-algebras.

If A is an algebra, then a morphism from A to itself is called an ***endomorphism*** of A.

We give examples of algebra morphisms.

I.4.2 Examples.

(a) Let A, B be K-algebras, with B a subalgebra of A. The inclusion map $\iota : B \to A$ defined by $b \mapsto b$, for $b \in B$, is a morphism, called the ***inclusion morphism*** or the ***injection morphism***. If $B = A$, this morphism is just the identity map on A (or B) denoted by 1_A. This notation is not to be confused with the multiplicative identity of A, because, as pointed out, the latter is now simply denoted by 1.

(b) Let A be a K-algebra and I an ideal of A. The map $\pi : A \to A/I$ given by $a \mapsto a + I$, for $a \in A$, is a morphism. Indeed, let $a, a_1, a_2 \in A$ and $\alpha \in K$, then

$$\begin{aligned} \pi(a_1 + a_2) &= (a_1 + a_2) + I = (a_1 + I) + (a_2 + I) = \pi(a_1) + \pi(a_2), \\ \pi(a_1 a_2) &= (a_1 a_2) + I = (a_1 + I)(a_2 + I) = \pi(a_1)\pi(a_2), \\ \pi(a\alpha) &= (a\alpha) + I = (a + I)\alpha = \pi(a)\alpha, \end{aligned}$$

because of the definition of the operations on A/I, while $\pi(1) = 1 + I$ follows from the definition of π. Actually, the previous equalities say more: we defined exactly those operations on A/I which make π an algebra morphism. The map π is called the ***projection*** or ***surjection morphism***.

(c) Let $\varphi : A \to B$ and $\psi : B \to C$ be algebra morphisms. Then so is their composite $\psi \circ \varphi : A \to C$.

(d) Let $\lambda \in K$ be fixed and define $\varphi : K[t] \to K$ as follows: for a polynomial $p = \sum_{i=0}^{d} \alpha_i t^i$, set $\varphi(p) = \sum_{i=0}^{d} \alpha_i \lambda^i$, that is, $\varphi(p)$ is the evaluation $p(\lambda)$ of the polynomial at λ. It follows from properties of evaluation that φ is a morphism: for instance, if $p, q \in K[t]$, then

$$\varphi(p+q) = (p+q)(\lambda) = p(\lambda) + q(\lambda) = \varphi(p) + \varphi(q)$$

and similarly for axioms (AM2) and (AM3). Finally, $\varphi(1) = 1$ because the 1 in $K[t]$ is the constant polynomial equal to 1.

Because algebra morphisms are ring homomorphisms which are also K-linear, they satisfy the usual properties of these maps. We recall some definitions. Let $\varphi : A \to B$ be a map. For any $A' \subseteq A$, its ***image*** is the subset $\varphi(A') = \{\varphi(a) \in B : a \in A'\}$, and the ***preimage*** of $B' \subseteq B$ is the subset $\varphi^{-1}(B') = \{a \in A : \varphi(a) \in B'\}$. The ***image*** $\operatorname{Im} \varphi$ of the morphism φ is $\varphi(A)$ and its ***kernel*** $\operatorname{Ker} \varphi$ is $\varphi^{-1}(0)$. We warn the reader that this notation does *not* mean that φ^{-1} exists as a map, it only does in case φ is bijective, see Lemma I.4.6 below. We list some properties of algebra morphisms.

I.4.3 Proposition. *Let $\varphi : A \to B$ be an algebra morphism. Then:*

(a) $\varphi(0) = 0$.
(b) $\varphi(-a) = -\varphi(a)$, *for any $a \in A$.*
(c) $\varphi(na) = n\varphi(a)$, *for any $a \in A, n \in \mathbb{Z}$.*
(d) $\varphi(a^n) = \varphi(a)^n$, *for any $a \in A$ and integer $n \geq 0$.*
(e) $\operatorname{Im} \varphi$ *is a subalgebra of B.*
(f) $\operatorname{Ker} \varphi$ *is an ideal of A.*
(g) φ *is injective if and only if* $\operatorname{Ker} \varphi = 0$.

Proof. (a), (b), (c), (g) are properties of abelian group homomorphisms and (d) a property of monoid homomorphisms. We prove (e) and (f). Because φ is a ring homomorphism, $\operatorname{Im} \varphi$ is a subring of B. It is actually a subalgebra of B, because, if $c \in \operatorname{Im} \varphi$ and $\alpha \in K$, then there exists $a \in A$ such that $c = \varphi(a)$, so $c\alpha = \varphi(a)\alpha = \varphi(a\alpha) \in \operatorname{Im} \varphi$. Similarly, $a \in \operatorname{Ker} \varphi$ implies $\varphi(ca) = 0 = \varphi(ac)$, for any $c \in A$, thus $\operatorname{Ker} \varphi$ is an ideal of A considered as ring. Then it is also an ideal of A viewed as algebra, see comments after Definition I.3.5. □

I.4.4 Example. Let K be a field, $\lambda \in K$ and $\varphi : K[t] \to K$ the evaluation at λ, see Example I.4.2(d) above. Then φ is surjective, that is, $\operatorname{Im} \varphi = K$, because, for any $\alpha \in K$, the constant polynomial equal to α evaluates (obviously!) to α. We show that $\operatorname{Ker} \varphi$ is the ideal $\langle t - \lambda \rangle$ generated by the polynomial $t - \lambda$. Because $K[t]$ is commutative, this ideal is the set of (polynomial) multiples of $t - \lambda$. Plainly, $(t - \lambda) \in \operatorname{Ker} \varphi$ and hence $\langle t - \lambda \rangle \subseteq \operatorname{Ker} \varphi$. Conversely, let $p \in \operatorname{Ker} \varphi$. Dividing p by $t - \lambda$, we get $p = (t - \lambda)q + r$ where q is the quotient and the remainder r is zero or of degree less than one, thus it is a constant. Evaluating both sides at λ yields $0 = p(\lambda) = (\lambda - \lambda)q(\lambda) + r$, because evaluation is an algebra morphism, and so $p \in \operatorname{Ker} \varphi$. Hence, $r = 0$ and $p = (t - \lambda)q \in \langle t - \lambda \rangle$. This completes the proof.

In all areas of algebra (and other fields of mathematics), there exists a notion of isomorphism, meant to express that two structures are 'identical except for the names of their elements'. This somewhat vague idea can be expressed rigorously.

I.4.5 Definition. A morphism of K-algebras $\varphi : A \to B$ is an ***isomorphism*** if there exists a morphism $\varphi' : B \to A$ such that $\varphi' \circ \varphi = 1_A$ and $\varphi \circ \varphi' = 1_B$. The algebras A and B are then said to be ***isomorphic***, which we denote as $A \cong B$.

The definition of isomorphism implies that it is invertible as a map, and hence bijective. Moreover, the morphism φ' of the definition is the uniquely determined inverse map of φ. We thus write φ^{-1} instead of φ'.

Clearly, the identity morphism $1_A : A \to A$ is an isomorphism. If $\varphi : A \to B$ is an isomorphism, then so is $\varphi^{-1} : B \to A$ and we have $(\varphi^{-1})^{-1} = \varphi$. Finally, if $\varphi : A \to B$, $\psi : B \to C$ are isomorphisms, then so is $\psi\varphi : A \to C$ and we have $(\psi\varphi)^{-1} = \varphi^{-1}\psi^{-1}$. In particular, the relation $\cong$ between algebras is reflexive, symmetric and transitive.

An isomorphism from an algebra A to itself is called an ***automorphism*** of A. The previous considerations imply that the set $\operatorname{Aut} A$ of automorphisms of A is a group under composition of morphisms. It is called the ***automorphism group*** of A.

Isomorphisms are bijective morphisms. The remarkable thing is that the converse also holds true: any bijective morphism is an isomorphism.

I.4.6 Lemma. *A morphism of algebras is an isomorphism if and only if it is bijective.*

Proof. We just need to prove sufficiency. Let $\varphi : A \to B$ be a bijective morphism. We must prove that the inverse map φ^{-1} is an algebra morphism. Let $b_1, b_2 \in B$, then:

$$\begin{aligned}
\varphi^{-1}(b_1 + b_2) &= \varphi^{-1}(1_B(b_1) + 1_B(b_2)) = \varphi^{-1}((\varphi \circ \varphi^{-1})(b_1) + (\varphi \circ \varphi^{-1})(b_2)) \\
&= \varphi^{-1}[\varphi(\varphi^{-1}(b_1)) + \varphi(\varphi^{-1}(b_2))] = (\varphi^{-1} \circ \varphi)[\varphi^{-1}(b_1) + \varphi^{-1}(b_2)] \\
&= 1_A[\varphi^{-1}(b_1) + \varphi^{-1}(b_2)] = \varphi^{-1}(b_1) + \varphi^{-1}(b_2)
\end{aligned}$$

where, at the fourth equality, we used that φ is a morphism. The other conditions are verified in the same way. □

Isomorphic algebras have the same properties. For instance, it is easily seen that if $A \cong B$, then A is commutative if and only if so is B.

In order to see examples of isomorphisms, recall that we used in Examples I.3.11(f) and (h) the expression 'is another realisation of': by this, we meant 'is isomorphic to'. The reader is encouraged to construct explicitly the isomorphism in each case.

I.4.2 The isomorphism theorems

It may be useful, in a given problem, to replace an algebra by another one which has the same properties inasmuch as our problem is concerned but is easier to handle. Hence the usefulness of the isomorphism theorems which show how to transfer information from one algebra to another.

We need a convention: if, in the sequel, a statement asserts the existence of a map completing a diagram, then this map is represented by dotted lines on the diagram (like the morphism $\bar{\varphi}$ in the theorem below). To say that a diagram consisting of objects (here, algebras) and maps (here, morphisms) is commutative, or commutes, means that if one composes maps on a path from an object to another, then the result of the composition depends only on these two objects, and does not depend on the path followed.

I.4.7 Theorem. *Let $\varphi : A \to B$ be an algebra morphism, $\iota : \operatorname{Im} \varphi \to B$ the inclusion and $\pi : A \to A/\operatorname{Ker} \varphi$ the projection. Then there exists a unique morphism $\bar{\varphi} : A/\operatorname{Ker} \varphi \to \operatorname{Im} \varphi$ which makes the following diagram commutative*

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d, "\pi"'] & B \\ A/\operatorname{Ker} \varphi \arrow[r, dashed, "\bar{\varphi}"] & \operatorname{Im} \varphi \arrow[u, "\iota"'] \end{tikzcd}$$

that is, such that $\varphi = \iota \circ \bar{\varphi} \circ \pi$. Moreover, $\bar{\varphi}$ is an isomorphism. In particular, $A/\operatorname{Ker} \varphi \cong \operatorname{Im} \varphi$.

Proof. We outline the strategy: we start by assuming that $\bar{\varphi}$ exists, and prove its uniqueness by finding a formula giving it explicitly. Then we use the same formula to prove its existence. Set for short $I = \operatorname{Ker} \varphi$. An arbitrary element of A/I is of the form $a + I = \pi(a)$, for some $a \in A$. One must then have

$$\bar{\varphi}(a + I) = \bar{\varphi}\pi(a) = \iota\bar{\varphi}\pi(a) = \varphi(a),$$

where the second equality holds true because $b \in B$ lies in $\operatorname{Im} \varphi$ if and only if $b = \iota(b)$. This shows uniqueness of $\bar{\varphi}$. But this formula also defines unambiguously a map, because if $a, a' \in A$ are such that $a + I = a' + I$, then $a - a' \in I = \operatorname{Ker} \varphi$ gives $\varphi(a - a') = 0$ and hence $\varphi(a) = \varphi(a')$. It is easy to verify that $\bar{\varphi}$ is an algebra morphism, and that it makes the shown diagram commutative.

Because the very definition of $\bar{\varphi}$ implies its surjectivity, we prove injectivity. Let $a \in A$ be such that $\bar{\varphi}(a+I) = 0$. Then $\varphi(a) = 0$ gives $a \in \operatorname{Ker} \varphi = I$ and so $a+I = I$, which is the zero of A/I. □

Theorem I.4.7 can be thought of as saying that, given an algebra morphism $\varphi : A \to B$, 'part' of A (more precisely, a quotient algebra of A) is the same as, that is, is isomorphic to, 'part' of B (more precisely, a subalgebra of B).

A situation which occurs frequently is when $\varphi : A \to B$ is surjective. In this case, $\operatorname{Im} \varphi = B$ and $\iota = 1_B$ so $\bar{\varphi}$ defines an isomorphism $A/\operatorname{Ker} \varphi \cong B$.

In Example I.4.4, we proved that if K is a field, then evaluation at $\lambda \in K$ induces a surjective algebra morphism from $K[t]$ to K whose kernel is the ideal $\langle t - \lambda \rangle$. Hence, $K[t]/\langle t - \lambda \rangle \cong K$.

On the other hand, if $\varphi : A \to B$ is injective, then $\operatorname{Ker} \varphi = 0$. In this case, A is isomorphic to a subalgebra of B.

The next isomorphism theorem compares two quotients of the same algebra.

I.4.8 Theorem. *Let A be an algebra, I, J ideals of A such that $I \subseteq J$ and $\pi_I : A \to A/I$, $\pi_J : A \to A/J$ the respective projections. Then there exists a unique morphism*

$\varphi : A/I \to A/J$ such that $\varphi \circ \pi_I = \pi_J$. Moreover, φ is surjective and its kernel is J/I.

$$\begin{tikzcd} & A \arrow[dl, "\pi_I"'] \arrow[dr, "\pi_J"] & \\ A/I \arrow[rr, dashed, "\varphi"] & & A/J \end{tikzcd}$$

Proof. We follow the same strategy of proof as in the previous theorem. An element of A/I is of the form $a + I = \pi_I(a)$, for some $a \in A$. One must then have

$$\varphi(a + I) = \varphi\pi_I(a) = \pi_J(a) = a + J.$$

This shows uniqueness. The same formula defines unambiguously a map, because if $a, a' \in A$ are such that $a + I = a' + I$, then $a - a' \in I \subseteq J$ gives $a + J = a' + J$. Again, one checks easily that φ is an algebra morphism.

Surjectivity of φ follows from its definition. One has $a + I \in \operatorname{Ker} \varphi$ if and only if $a + J = J$, that is, $a \in J$, so that $\operatorname{Ker} \varphi = \{a + I : a \in J\} = J/I$, as required. □

As a consequence, J/I is an ideal of the algebra A/I, because of Proposition I.4.3(f), and then Theorem I.4.7 gives an algebra isomorphism $(A/I)/(J/I) \cong A/J$.

The third isomorphism theorem compares quotients of subalgebras of a given algebra. Let A be an algebra, B a subalgebra and I an ideal of A. Using Lemma I.3.4, one proves easily that $B + I = \{b + x : b \in B, x \in I\}$ is a subalgebra of A, in which I is an ideal. Thus, the quotient $(B + I)/I$ makes sense (and is an algebra).

I.4.9 Theorem. *Let A be an algebra, B a subalgebra and I an ideal of A. Then there exists an algebra isomorphism*

$$\frac{B}{B \cap I} \cong \frac{B + I}{I}.$$

Proof. Our strategy here consists in constructing a surjective algebra morphism $\varphi : B \to (B + I)/I$ whose kernel is $B \cap I$, then applying Theorem I.4.7.

Set $\varphi : b \mapsto b + I$, for $b \in B$. Then φ is easily seen to be a morphism. It is surjective because an arbitrary element of $(B + I)/I$ is of the form $(b + x) + I$, for some $b \in B, x \in I$, and we have $b + x + I = b + I = \varphi(b)$, because $x \in I$. The kernel consists of all $b \in B$ such that $b + I = I$ or, equivalently, $b \in I$: this is $B \cap I$, as required. Theorem I.4.7 yields the result. □

In particular, $B \cap I$ is an ideal of B.

This theorem is called the 'parallelogram law': it asserts that the notched opposite sides of the following parallelogram are isomorphic.

$$\begin{tikzcd} & B + I \arrow[dl, no head] \arrow[dr, no head] & \\ B \arrow[dr, no head] & & I \arrow[dl, no head] \\ & B \cap I & \end{tikzcd}$$

Here, $B \cap I$ is the largest ideal (in fact, the largest subset) contained in both B and I. On the other hand, $B + I$ is the smallest abelian group containing both B and I, and it is a subalgebra of A, hence it is the smallest subalgebra containing both B and I. The segment (side) from $B \cap I$ to B represents the quotient $B/(B \cap I)$. Similarly, the opposite segment represents the quotient $(B + I)/I$. The similar notches mean that they are isomorphic. The other two (unnotched) sides of the parallelogram do not correspond to algebras: they are just K-modules.

We finish this subsection with a theorem comparing the ideals in an algebra and one of its quotients. For this reason, it is called the ***correspondence of ideals theorem***. Recall that if $\varphi : A \to B$ is a map and $A' \subseteq A$, then one always has $A' \subseteq \varphi^{-1}(\varphi(A'))$ and equality holds true when φ is injective. Also, if $B' \subseteq B$, then one has $\varphi(\varphi^{-1}(B')) \subseteq B'$ and equality holds true when φ is surjective.

I.4.10 Theorem. *Let A be an algebra, I an ideal and $\pi : A \to A/I$ the projection. The map $J \mapsto \pi(J)$ is an inclusion-preserving bijection from the set of ideals of A containing I to the set of ideals of A/I. The inverse bijection is $\bar{J} \mapsto \pi^{-1}(\bar{J})$.*

Proof. Let J be an ideal of A containing I. Because of Theorem I.4.8, $\pi(J) = J/I = \{x + I : x \in J\}$ is an ideal of A/I. Clearly, if $J_1 \subseteq J_2$, then $\pi(J_1) \subseteq \pi(J_2)$. Moreover, $J \subseteq \pi^{-1}\pi(J)$. We show the reverse inclusion. An element $a \in A$ lies in $\pi^{-1}\pi(J)$ if and only if $\pi(a) \in \pi(J)$, that is, there exists $x \in J$ such that $\pi(a) = \pi(x)$, or, equivalently, $a - x \in \operatorname{Ker}\pi = I \subseteq J$. So, $a \in J$ and hence $J = \pi^{-1}\pi(J)$. On the other hand, if $\bar{J}$ is an ideal in A/J, then $\pi^{-1}(\bar{J}) = \{a \in A : a + I \in \bar{J}\}$ is an ideal of A containing $I = \operatorname{Ker}\pi$, see Exercise I.4.4(d). Finally, $\pi\pi^{-1}(\bar{J}) = \bar{J}$, because π is surjective. The proof is complete. □

As a corollary, we characterise maximal ideals in commutative algebras. A proper ideal I in an algebra A is called ***maximal*** if no ideal can be properly inserted between I and A, that is, if J is an ideal such that $I \subseteq J \subseteq A$, then $J = I$ or $J = A$.

I.4.11 Corollary. *Let A be a nonzero commutative algebra. A proper ideal I in A is maximal if and only if A/I is a field.*

Proof. Because A is commutative, so is A/I. Hence, according to Proposition I.3.9, A/I is a field if and only if it is nonzero and has no nonzero proper ideal. Because of Theorem I.4.10, this happens if and only if $A \neq I$ and no ideal is properly contained between A and I, that is, I is a maximal ideal. □

For instance, if K is a field and $\lambda \in K$, then $K[t]/\langle t - \lambda\rangle \cong K$, see the example just after Theorem I.4.7. Hence, the ideal $\langle t - \lambda\rangle$ is maximal in $K[t]$, for any λ.

Exercises of Section I.4

1. Let $\varphi : A \to B, \varphi' : A' \to B'$ be algebra morphisms. Define $\Phi : A \times A' \to B \times B'$ by $(a, a') \mapsto (\varphi(a), \varphi'(a'))$, for $a \in A, a' \in A'$. Prove that:

(a) Φ is an algebra morphism.
(b) $\operatorname{Ker}\Phi = \operatorname{Ker}\varphi \times \operatorname{Ker}\varphi'$.
(c) $\operatorname{Im}\Phi = \operatorname{Im}\varphi \times \operatorname{Im}\varphi'$.

2. Given a non necessarily unitary algebra A, denote by A_1 the algebra obtained from A by adjoining an identity, see Exercise I.3.1.
(a) Prove that, if A is already unitary, then A_1 is isomorphic to $A \times K$. In particular, A_1 is not the smallest unitary algebra containing A.
(b) Let $\varphi : A \to B$ be a morphism of nonunitary algebras, that is, a map such that $\varphi(a + a') = \varphi(a) + \varphi(a')$, $\varphi(a\alpha) = \varphi(a)\alpha$ and $\varphi(aa') = \varphi(a)\varphi(a')$, for $a, a' \in A$ and $\alpha \in K$. Prove that φ uniquely extends to a morphism $\varphi_1 : A_1 \to B_1$.
(c) Prove that, conversely, any morphism $\varphi_1 : A_1 \to B_1$ restricts to a morphism $\varphi : A \to B$.
(d) Prove that $A \cong B$ if and only if $A_1 \cong B_1$.

3. Let K be a field and A a finite-dimensional K-algebra with basis $\{e_1, \cdots, e_n\}$.
(a) Prove that A is uniquely determined, up to isomorphism, by a family of constants $(\alpha_{ij}^k)_{1 \leqslant i,j,k \leqslant n}$ from K defined by $e_i e_j = \sum_{k=1}^n \alpha_{ij}^k e_k$. The α_{ij}^k are called the ***structure constants*** of the algebra.
(b) Deduce that, if K is a finite field, then there are, up to isomorphism, only finitely many algebras of a given dimension n.
(c) Prove that associativity of multiplication in A is expressed by the relations $\sum_{\ell=1}^n \alpha_{ij}^\ell \alpha_{\ell k}^m = \sum_{\ell=1}^n \alpha_{jk}^\ell \alpha_{i\ell}^m$, for all i, j, k, ℓ, m (so, structure constants cannot be chosen arbitrarily).

4. Let $\varphi : A \to B$ be an algebra morphism. Prove the following:
(a) For any subalgebra A' of A, its image $\varphi(A')$ is a subalgebra of B.
(b) For any subalgebra B' of B, its preimage $\varphi^{-1}(B')$ is a subalgebra of A.
(c) For any ideal I of A, its image $\varphi(I)$ is an ideal of $\operatorname{Im}\varphi$. Is it necessarily an ideal of B?
(d) For any ideal J of B, its preimage $\varphi^{-1}(J)$ is an ideal of A containing $\operatorname{Ker}\varphi$.

5. Let A, B be finite-dimensional algebras over the same field K. If $(e_i)_{1 \leqslant i \leqslant n}$ is a basis of A, and $\varphi : A \to B$ a K-linear map such that $(\varphi(e_i))_{1 \leqslant i \leqslant n}$ is a basis of B, then φ is an algebra isomorphism if and only if $\varphi(e_i)\varphi(e_j) = \varphi(e_i e_j)$, for any i, j.

6. Let A be an algebra containing a field K as a subalgebra, B an algebra and $\varphi : A \to B$ an injective algebra morphism. Prove that B contains a field isomorphic to K.

7. Prove that the matrix algebra of Exercise I.2.9 is isomorphic to the $\mathbb{R}$-algebra of the quaternions.

8. Given a quiver $Q = (Q_0, Q_1, s, t)$, its ***opposite quiver*** $Q^{\mathrm{op}} = (Q_0', Q_1', s', t')$ is obtained from Q by reversing the direction of arrows, that is, it is defined as follows: $Q_0' = Q_0$, $Q_1' = Q_1$ and, if $\alpha \in Q_1$, then $s'(\alpha) = t(\alpha)$, $t'(\alpha) = s(\alpha)$. Let K be a field and Q a finite quiver. Prove that:
(a) $(KQ)^{\mathrm{op}} \cong K(Q^{\mathrm{op}})$.
(b) If I is an admissible ideal in KQ, then there exists an admissible ideal I^{op} in KQ^{op} such that $KQ^{\mathrm{op}}/I^{\mathrm{op}} \cong (KQ/I)^{\mathrm{op}}$.

9. Prove that the map $z \mapsto \bar{z}$ is an automorphism of $\mathbb{C}$ considered as an $\mathbb{R}$-algebra.

10. Let K be a field and $n > 0$ an integer. Prove that we have isomorphisms between:

(a) The centre of the full matrix algebra $M_n(K)$ and K.

(b) The incidence algebra of the poset $E = \{1, 2, 3\}$, ordered by $1 \leqslant 3, 2 \leqslant 3$, and the subalgebra of $T_3(K)$ given by

$$\begin{pmatrix} K & 0 & 0 \\ 0 & K & 0 \\ K & K & K \end{pmatrix} = \left\{ \begin{pmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \\ \alpha_3 & \alpha_4 & \alpha_5 \end{pmatrix} : \alpha_i \in K \right\}.$$

(c) $K[s, t]/\langle t - s\rangle$ and $K[t]$.

(d) The path algebra of the quiver

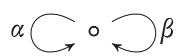
$$n \to n-1 \to \cdots \to 2 \to 1$$

and the lower triangular matrix algebra $T_n(K)$.

(e) The path algebra of the one-loop quiver

bound by the ideal $\langle \alpha^n \rangle$ and $K[t]/\langle t^n \rangle$.

(f) The path algebra of the two-loops quiver



bound by the ideal $\langle \alpha\beta - \beta\alpha \rangle$ and $K[s, t]$.

(g) The path algebra of the quiver

$$\begin{tikzcd} & \circ \arrow[dl, "\beta"'] & \\ \circ & & \circ \arrow[ul, "\alpha"'] \arrow[dl, "\gamma"] \\ & \circ \arrow[ul, "\delta"] & \end{tikzcd}$$

bound by the ideal $\langle \alpha\beta - \gamma\delta \rangle$ and the subalgebra of $T_4(K)$ given by

$$\begin{pmatrix} K & 0 & 0 & 0 \\ K & K & 0 & 0 \\ K & 0 & K & 0 \\ K & K & K & K \end{pmatrix} = \left\{ \begin{pmatrix} \alpha_1 & 0 & 0 & 0 \\ \alpha_2 & \alpha_3 & 0 & 0 \\ \alpha_4 & 0 & \alpha_5 & 0 \\ \alpha_6 & \alpha_7 & \alpha_8 & \alpha_9 \end{pmatrix} : \alpha_i \in K \text{ for all } i \right\}.$$

(h) The path algebra of the quiver

$$\circ \leftleftarrows \circ$$

and the algebra of lower triangular matrices

$$\begin{pmatrix} \alpha & 0 & 0 \\ \beta & \gamma & 0 \\ \delta & 0 & \gamma \end{pmatrix}$$

with $\alpha, \beta, \gamma, \delta \in K$ and the usual matrix operations.

11. Let K be a field. Consider the set of matrices

$$A = \begin{pmatrix} K & 0 & 0 \\ K & K & 0 \\ 0 & K & K \end{pmatrix} = \left\{ \begin{pmatrix} \alpha_1 & 0 & 0 \\ \alpha_2 & \alpha_3 & 0 \\ 0 & \alpha_4 & \alpha_5 \end{pmatrix} : \alpha_i \in K \right\}.$$

(a) Prove that A becomes an algebra, when given the usual K-vector space structure of $T_3(K)$ and the multiplication defined by

$$\begin{pmatrix} \alpha_1 & 0 & 0 \\ \alpha_2 & \alpha_3 & 0 \\ 0 & \alpha_4 & \alpha_5 \end{pmatrix} \cdot \begin{pmatrix} \beta_1 & 0 & 0 \\ \beta_2 & \beta_3 & 0 \\ 0 & \beta_4 & \beta_5 \end{pmatrix} = \begin{pmatrix} \alpha_1\beta_1 & 0 & 0 \\ \alpha_2\beta_1 + \alpha_3\beta_2 & \alpha_2\beta_2 & 0 \\ 0 & \alpha_4\beta_3 + \alpha_5\beta_4 & \alpha_3\beta_3 \end{pmatrix}.$$

for $\alpha_i, \beta_i \in K$ for all i.

(b) Prove that $A \cong T_3(K)/I$, where $I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ K & 0 & 0 \end{pmatrix}$.

12. Let A be a K-algebra. Prove that:

(a) The set $\operatorname{End}_K A$ admits a natural K-algebra structure.

(b) Given $a \in A$, the map $\rho_a : A \to A$ defined by $x \mapsto xa$, for $x \in A$, is a K-linear endomorphism of A.

(c) The map $a \mapsto \rho_a$, for $a \in A$, is an injective algebra morphism from A to $\operatorname{End}_K A$.

(d) If K is a field, then any finite-dimensional algebra is a subalgebra of a full matrix algebra.

13. Let $n > 0$ be an integer. Construct a surjective algebra morphism from $T_n(K)$ to K^n and compute its kernel.

14. Let A be an algebra, I an ideal and $\pi : A \to A/I$ the projection. Prove that, for any $J \trianglelefteq A$, we have $\pi(J) = (I + J)/I$.

15. Let A be an algebra and I an ideal. Prove that:

(a) $M_n(I)$ is an ideal in the algebra $M_n(A)$.

(b) $M_n(A)/M_n(I) \cong M_n(A/I)$.

16. Prove that the ideal $\langle t^2 + 1 \rangle$ is maximal in $\mathbb{R}[t]$.

17. Let d be a positive integer which is not a perfect square. Prove that:

(a) $\mathbb{Q}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Q}\}$ is a subalgebra of $\mathbb{R}$, and is a two-dimensional $\mathbb{Q}$-algebra.

(b) The map $a + b\sqrt{d} \mapsto a - b\sqrt{d}$, for $a, b \in \mathbb{Q}$, is an automorphism of $\mathbb{Q}[\sqrt{d}]$ as a $\mathbb{Q}$-algebra.

(c) $A = \left\{ \begin{pmatrix} a & b \\ db & a \end{pmatrix} : a, b \in \mathbb{Q} \right\}$, with usual matrix operations, is a subalgebra of $M_2(\mathbb{Q})$.

(d) The algebra A of (c) above is isomorphic to $\mathbb{Q}[\sqrt{d}]$.

(e) $\mathbb{Q}[\sqrt{2}]$ and $\mathbb{Q}[\sqrt{3}]$ are not isomorphic. Can one deduce a necessary and sufficient condition on two positive integers d, d' so that $\mathbb{Q}[\sqrt{d}] \cong \mathbb{Q}[\sqrt{d'}]$?

18. Let A be a commutative algebra. An ideal P of A is called a ***prime ideal*** if $ab \in P$ implies $a \in P$ or $b \in P$. Prove that:
   (a) An ideal in $\mathbb{Z}$ is prime if and only if it is generated by a prime integer.
   (b) An ideal P in A is prime if and only if A/P is an integral domain.
   (c) Any maximal ideal is prime.
   (d) The ideal $\langle t^2 + 1 \rangle$ is prime but not maximal in $\mathbb{Z}[t]$.

19. This exercise outlines another proof of Theorem I.4.9. Let A be an algebra, B a subalgebra and I an ideal.
   (a) Prove that $B + I = \{b + x : b \in B, x \in I\}$ is a subalgebra of A, and in fact the smallest subalgebra containing B and I.
   (b) Construct a surjective algebra morphism $\psi : B+I \to B/(B \cap I)$ with kernel I.
   (c) Deduce Theorem I.4.9.

20. Let $\varphi : A \to B$ be a surjective algebra morphism. Prove that:
   (a) If I is an ideal of A containing Ker φ, then $A/I \cong B/\varphi(I)$.
   (b) If J is an ideal of B, then $B/J \cong A/\varphi^{-1}(J)$.

21. Let $\varphi : A \to B$ be an algebra morphism, $I \trianglelefteq A$ and $J \trianglelefteq B$. Denote by $\pi_I : A \to A/I, \pi_J : B \to B/J$ the respective projections. Prove that $\varphi(I) \subseteq J$ if and only if there exists a unique morphism $\bar{\varphi} : A/I \to B/J$ such that $\bar{\varphi} \circ \pi_I = \pi_J \circ \varphi$. If this is the case, then compute the kernel and the image of $\bar{\varphi}$.

22. Let K be a field of characteristic two, see Exercise I.2.5, and $G = \{1, g, h, gh\}$ the Klein Four-group. Define $\varphi : K[s,t] \to KG$ by $\varphi(s) = 1 + g$, $\varphi(t) = 1 + h$.
   (a) Prove that φ is surjective of kernel $\langle s^2, t^2 \rangle$.
   (b) Deduce that $KG \cong K[s,t]/\langle s^2, t^2 \rangle$.
   (c) Let Q be the two-loops quiver

$$\begin{tikzcd} \circ \arrow[loop left, "\alpha"] \arrow[loop right, "\beta"] \end{tikzcd}$$

   Prove that $KG \cong KQ/\langle \alpha^2, \beta^2, \alpha\beta - \beta\alpha \rangle$.

I.5 Principal ideal domains

I.5.1 Irreducible elements

The first examples of rings one encounters are usually the ring $\mathbb{Z}$ of integers and the ring $K[t]$ of polynomials in t with coefficients in a field K. Both are integral domains which share the additional property that in each, one can perform efficiently a division. They also share another, less evident, property: in both, any ideal of the ring is principal, that is, is generated by a single element. This leads to the following definition.

I.5.1 Definition. An algebra (or a ring) A is a ***principal ideal domain*** if it is an integral domain in which any ideal is principal, that is, is of the form $\langle a \rangle = Aa$, for some $a \in A$.

In particular, principal ideal domains are commutative. Any field K is a principal ideal domain because it has only two ideals: $0 = \langle 0 \rangle$ and $K = \langle 1 \rangle$. We already observed after Lemma I.3.7 that $\mathbb{Z}$ is a principal ideal domain.

The next examples use the division of polynomials, that we recall here. Let A be an integral domain, then the algebra $A[t]$ of polynomials in t with coefficients in A is an integral domain, see the comments after Definition I.2.4. The division theorem for polynomials asserts that if $f, g \in A[t]$ are such that the leading coefficient of g is invertible (in particular, $g \neq 0$), then there exist unique polynomials q (the ***quotient***) and r (the ***remainder***) such that $f = gq + r$ and either $r = 0$ or the degree $\deg(r)$ of r is strictly less than the degree $\deg(g)$ of g. In particular, division by a nonzero polynomial is always possible if $A = K$ is a field.

I.5.2 Examples.

(a) Recall that the ***Well-Ordering Principle*** states that any set can be ordered so that any of its nonempty subsets has a least element. The Well-Ordering Principle is equivalent to the Axiom of Choice (hence to Zorn's lemma).

A polynomial algebra $K[t]$ with coefficients in a *field* K is a principal ideal domain. Let I be a nonzero ideal in $K[t]$. Because $I \neq 0$, the set $\{\deg(f) : f \neq 0, f \in I\}$ is a nonempty set of natural numbers. Because of the Well-Ordering Principle, there exists in I a nonzero polynomial p of lesser degree. Clearly, $\langle p \rangle \subseteq I$. Conversely, let $f \in I$. Because of the division theorem, and because $p \neq 0$, there exist $q, r \in K[t]$ such that $f = pq + r$ and $r = 0$ or $\deg(r) < \deg(p)$. Because $f, p \in I$, we have $r = f - qp \in I$. But p is of lesser degree in I. Hence, $r = 0$ and $f = qp \in \langle p \rangle$. Therefore, $I = \langle p \rangle$, as required.

(b) On the other hand, a polynomial algebra $K[t]$ with coefficients in a *ring* K (even a principal ideal domain) is usually not a principal ideal domain. For instance, $\mathbb{Z}[t]$ is an integral domain, but not a principal ideal domain. Indeed, division by a nonzero polynomial is not always possible in $\mathbb{Z}[t]$, such as, for instance, division by an integer $n > 1$. Consider the ideal $\langle 2, t \rangle$ generated by 2 and t in $\mathbb{Z}[t]$. If it is principal and generated by p, say, then p divides both 2 and t. Hence, $p \in \{-1, +1\}$. But $\langle 1 \rangle = \langle -1 \rangle = \mathbb{Z}[t] \neq \langle 2, t \rangle$, a contradiction. Therefore, $\langle 2, t \rangle$ is not a principal ideal.

The proof in example (a) above rests on the notion of degree of a polynomial. One can generalise this notion by defining a class of algebras, called ***Euclidean domains***, in which division is possible, and there is an integer-valued function, called ***valuation***, which behaves like the degree. Examples are the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, see Exercise I.2.8, the ring $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ and others, a proof is outlined in Exercise I.5.8 below.

We need considerations on division in integral domains. Let A be an integral domain and $a, b \in A$. Then b is said to ***divide*** a, in symbols $b \mid a$, if there exists $q \in A$ such that $a = qb$. The element q is the ***quotient*** of a by b. If $b \neq 0$, then, because of Lemma I.2.5, q is unique.

One word of caution: when we say that b divides a, it does *not* mean that we can multiply a by the multiplicative inverse of b. Indeed, inverses usually do not exist in integral domains. For instance, one has $0 \mid 0$, which means that 0 is a multiple of 0, because, for any $q \in A$, one has $0 = q \cdot 0$. Actually, for any $a \in A$, we have $1 \mid a$ and $a \mid 0$.

I.5.3 Lemma. *Let A be an integral domain and $a, b \in A$ nonzero. Then:*

(a) *$Aa \subseteq Ab$ if and only if $b \mid a$.*

(b) *$Aa = Ab$ if and only if there exists $u \in A$ invertible such that $a = ub$.*

Proof. (a) $Aa \subseteq Ab$ happens if and only if $a \in Ab$, that is, if and only if there exists $q \in A$ such that $a = qb$. This amounts to saying that $b \mid a$.

(b) Assume that $Aa = Ab$. Because of (a), we have $a \mid b$ and $b \mid a$, so there exist q, q' such that $b = qa$ and $a = q'b$. Then $a = qq'a$ and $a \neq 0$ yield $qq' = 1$, because of Lemma I.2.5. Therefore, q and q' are invertible. Conversely, if u is invertible and such that $a = ub$, then $b \mid a$ implies $Aa \subseteq Ab$. But u is invertible. So $b = u^{-1}a$ and $Ab \subseteq Aa$ as well. □

Consequently, $u \in A$ is invertible if and only if $Au = A$, or, equivalently, if and only if $u \mid 1$. If $a, b \in A$ are such that $Aa = Ab$, then they are called ***associate***. Equivalently, a, b are associate if and only if $a \mid b$ and $b \mid a$, or, if and only if there exists $u \in A$ invertible such that $a = ub$.

For example, if K is a field, an element of $K[t]$ is invertible if and only if it is a nonzero element of K. Thus, two polynomials $p, q \in K[t]$ are associate if and only if there exists $c \in K, c \neq 0$ such that $q = cp$. The consequence is that any $p \in K[t]$ is associate to a unique ***normalised*** polynomial that is, such that its leading coefficient equals $+1$.

It is well-known that any pair of integers, or of polynomials, has a greatest common divisor and a least common multiple. This property holds true in any principal ideal domain. We define these concepts.

I.5.4 Definition. Let A be a principal ideal domain and $a, b \in A$.

(a) A ***greatest common divisor*** of a, b, abbreviated to gcd, is an element $d \in A$ such that:

(i) $d \mid a$ and $d \mid b$, and

(ii) if $c \mid a$ and $c \mid b$, then $c \mid d$.

(b) A ***least common multiple*** of a, b, abbreviated to lcm, is an element $m \in A$ such that:

(i) $a \mid m$ and $b \mid m$, and

(ii) if $a \mid c$ and $b \mid c$, then $m \mid c$.

If it exists, a gcd is unique up to associates. For, if d, d' are both gcd of a, b, then, because d' is a common divisor of a, b, applying the definition of gcd to d yields $d \mid d'$. Similarly, $d' \mid d$. Hence, d and d' are associate. By abuse of notation (because d is not unique), we denote by $d = (a, b)$ a gcd of a, b.

In the same way, we can prove that an lcm m of a, b is unique up to associates, and, by abuse of notation, we write $m = [a, b]$ for an lcm of a, b, For instance, for any $a \in A$, we have $(a, 0) = a$ and $(a, 1) = 1$. Also, $[a, 0] = 0$ and $[a, 1] = a$. We prove our existence theorem for gcd and lcm.

I.5.5 Theorem. *Any pair of elements in a principal ideal domain has a gcd and an lcm.*

Proof. Let A be a principal ideal domain and $a, b \in A$. The sum and the intersection of ideals are ideals. Hence, there exist $d, m \in A$ such that $Aa + Ab = Ad$ and $Aa \cap Ab = Am$. It is easy to verify that d and m satisfy respectively the definition of a gcd and an lcm of a, b. $\square$

I.5.6 Corollary. *Let A be a principal ideal domain and $a, b \in A$. Then:*

(a) *If $(a, b) = d$, then there exist $s, t \in A$ such that $sa + tb = d$.*
(b) *$(a, b) = 1$ if and only if there exist $s, t \in A$ such that $sa + tb = 1$.*

Proof. (a) Because of the proof of Theorem I.5.5, a gcd d of a, b belongs to the ideal $Aa + Ab$. Hence, it can be written in the form $d = sa + tb$, for some $s, t \in A$.

(b) Necessity follows from (a). Conversely, let s, t be such that $sa + tb = 1$. We certainly have $1 \mid a$ and $1 \mid b$. If $d' \mid a$ and $d' \mid b$, then $d' \mid sa + tb = 1$. So d' is invertible and $(a, b) = 1$. $\square$

Two elements having 1 as a gcd are called ***coprime***. Corollary I.5.6(b) is a criterion allowing to verify whether two elements are coprime or not. It is known as the ***Bézout–Bachet theorem***.

We arrived at the main result of this subsection.

I.5.7 Theorem. *Let A be a principal ideal domain and p a nonzero element of A. The following conditions are equivalent:*

(a) *p is not invertible and $p \mid ab$ implies $p \mid a$ or $p \mid b$.*
(b) *p is not invertible and $p = ab$ implies that a or b is invertible.*
(c) *Ap is a maximal ideal in A.*

Proof. (a) implies (b). Assume that $p = ab$. Then $p \mid ab$. Consequently, $p \mid a$ or $p \mid b$. Assume the former, then there exists q such that $a = qp$. Then $p = ab = pqb$ and $p \neq 0$ yield $qb = 1$, because of Lemma I.2.5. Hence, b is invertible. Similarly, if $p \mid b$, then a is invertible.

(b) implies (c). Let $I \trianglelefteq A$ be such that $Ap \subseteq I \subseteq A$. Because A is principal, there exists $a \in A$ such that $I = Aa$. Because $Ap \subseteq Aa$, there exists $b \in A$ such that $p = ab$. The hypothesis (b) says that a or b is invertible. If a is invertible, then, because of Lemma I.5.3, we have $I = Aa = A$. If b is invertible, then, for the same reason, $I = Aa = Ap$.

(c) implies (a). Assume that Ap is maximal. Then $Ap \neq A$ shows that p is not invertible. If $p \mid ab$, then $ab \in Ap$, and so, in the quotient A/Ap, we have $(a + Ap)(b + Ap) = ab + Ap = 0$. Because Ap is maximal, it follows from

Corollary I.4.11 that A/Ap is a field. Hence, $a + Ap = 0$ or $b + Ap = 0$. Thus, $a \in Ap$ or $b \in Ap$. Equivalently, $p \mid a$ or $p \mid b$. □

Elements satisfying the equivalent conditions of the theorem are called ***irreducible elements*** in the principal ideal domain. Because of Lemma I.5.3, an element $p \in A$ is irreducible if and only if, for any invertible element $u \in A$, the element up is also irreducible, that is, if and only if the associates of p are irreducible. In this case, we call p and up ***associate irreducible*** elements. It follows from condition (a) of the theorem and induction that, if p is irreducible and $p \mid a_1 \cdots a_m$, for any integer $m > 1$, and the a_i elements of A, then there exists i such that $p \mid a_i$.

It is reasonable to ask whether an integral domain has 'enough' irreducible elements. Because of Theorem I.5.7, this is equivalent to the existence of 'enough' maximal ideals. We start by proving that maximal ideals exist in any nonzero algebra. This is known as ***Krull's theorem***.

I.5.8 Theorem. Krull's Theorem. *Let I_0 be a proper ideal in a nonzero algebra A. Then there exists a maximal ideal of A containing I_0. In particular, any nonzero algebra has maximal ideals.*

Proof. Let $\mathcal{E}$ be the set of proper ideals of A containing I_0. Because $I_0 \in \mathcal{E}$, we have $\mathcal{E} \neq \emptyset$. The set $\mathcal{E}$ is partially ordered by inclusion. We show that it is inductive. Let $\mathcal{F}$ be a chain in $\mathcal{E}$ and set $I = \cup_{J \in \mathcal{F}} J$. Then I is an ideal of A, see Exercise I.3.8, and contains I_0. Assume that $I = A$, then $A = \cup_{J \in \mathcal{F}} J$ gives $1 \in J$, for some $J \in \mathcal{F}$, and hence J equals A, which contradicts $J \in \mathcal{F} \subseteq \mathcal{E}$. Hence, I is a proper ideal and $I \in \mathcal{E}$. So, $\mathcal{E}$ is an inductive poset. Applying Zorn's lemma finishes the proof. The last statement follows by taking I_0 equal to the zero ideal. □

We explain what we meant by saying that a principal ideal domain has 'enough' irreducible elements.

I.5.9 Corollary. *Let A be a principal ideal domain. For any $a \in A$ which is neither zero nor invertible, there exists an irreducible element p such that $p \mid a$.*

Proof. Because a is not invertible, Aa is a proper ideal of A. Because of Krull's Theorem I.5.8, there exists a maximal ideal I containing Aa. But A is principal. Hence, there exists $p \in A$ such that $I = Ap$. Because I is maximal, p is irreducible, due to Theorem I.5.7. Because $Aa \subseteq Ap$, we have $p \mid a$. □

For instance, the irreducible elements in $\mathbb{Z}$ are the prime integers, as follows from Example I.2.8(b) and Corollary I.4.11. In $K[t]$, with K a field, we have seen at the end of Section I.4 that any polynomial of the form $t - \lambda$, for some λ, is irreducible. In general, polynomials of this form are not the only irreducible ones. For instance, Exercise I.4.18(c) says that $t^2 + 1$ is irreducible in $\mathbb{R}[t]$. The Fundamental Theorem of Algebra says that the only irreducible polynomials in $\mathbb{C}[t]$ are of the form $t - \lambda$, for some $\lambda \in \mathbb{C}$, and their associates, that is, their multiples by nonzero complex numbers.

I.5.10 Example. We apply Example I.5.2(a) above to algebraically closed fields. A commutative ring K is called ***algebraically closed*** if, for any nonconstant polynomial $p \in K[t]$, there exists $\alpha \in K$ such that $p(\alpha) = 0$. Then, α is called a ***root*** of p. Algebraically closed rings are actually fields: indeed, for any nonzero $\alpha \in K$, the polynomial $p = \alpha t - 1$ has a root, hence α has an inverse. The Fundamental Theorem of Algebra asserts that the field $\mathbb{C}$ of complex numbers is algebraically closed. One can prove that any field has an algebraically closed field extension.

Let D be a division ring extension of a field K, see Definition I.2.9, which is finite-dimensional as K-vector space. Then, for any $a \in D$, the infinite sequence of elements $\{1, a, \cdots, a^i, \cdots\}$ cannot be linearly independent over K. Hence, there exist $d > 0$ and $\alpha_0, \cdots, \alpha_d \in K$ such that $\alpha_0 + \alpha_1 a + \cdots + \alpha_d a^d = 0$, that is, the polynomial $p = \alpha_0 + \alpha_1 t + \cdots + \alpha_d t^d$ has a as a root. Therefore, the set of polynomials in $K[t]$ having a as a root is nonzero. Because this set is easily seen to be an ideal, there exists a unique normalised polynomial $m_a \in K[t]$ which generates it. Because m_a is of lesser degree, due to Example I.5.2(a), it is irreducible. It is called the ***minimal polynomial*** of a.

If K is algebraically closed, then no proper division ring extension D is finite-dimensional as K-vector space. For, let $a \in D$ and m_a its minimal polynomial. Because K is algebraically closed, the irreducible polynomial m_a is of degree one, hence of the form $m_a = t - a$. Therefore, $a \in K$. Thus, $D = K$.

I.5.2 The Unique Factorisation Theorem

The reader is certainly familiar with the so-called Fundamental Theorem of Arithmetics which asserts that any positive integer distinct from 1 decomposes uniquely as product of positive prime (=irreducible) integers, and this decomposition is unique up to order of the factors. This statement easily extends to any nonzero integer distinct from ± 1. A similar one holds true for polynomials over a field. Integers and polynomials are our main examples of principal ideal domains. We now prove that this unique factorisation property holds true in any principal ideal domain. We need a lemma (which will take its real significance in Chapter VII).

I.5.11 Lemma. *Let $(I_i)_{i \geq 0}$ be a sequence of ideals in a principal ideal domain A such that, for any $i \geq 0$, we have $I_i \subseteq I_{i+1}$. Then there exists $j \geq 0$ such that $I_j = I_{j+1} = I_{j+2} = \cdots$.*

Proof. The set $I = \cup_{i \geq 0} I_i$ is an ideal in A, see Exercise I.3.8. Because A is a principal ideal domain, there exists $a \in A$ such that $I = Aa$. But then there exists $j \geq 0$ such that $a \in I_j$. Hence, $I = Aa \subseteq I_j$, so that $I = I_j$. Therefore, $I_j = I_{j+k}$, for any $k \geq 0$. $\square$

We prove the Unique Factorisation Theorem.

I.5.12 Theorem. *Let A be a principal ideal domain and $a \in A$ nonzero and noninvertible. Then:*

(a) *a can be written in the form $a = p_1 \cdots p_m$, where each p_i is irreducible, and*
(b) *this factorisation is essentially unique: if $p_1 \cdots p_m = q_1 \cdots q_n$, with the p_i, q_j irreducible, then $m = n$ and there exists a permutation σ of $\{1, \cdots, m\}$ such that, for each i, p_i and $q_{\sigma(i)}$ are associate.*

Proof. (a) *Existence.* Because of Corollary I.5.9, there exists p_1 irreducible such that $p_1 \mid a$. Let a_1 be such that $a = p_1 a_1$. Then $a_1 \neq 0$. Inductively, we may assume that we have a factorisation $a = p_1 \cdots p_k\, a_k$, where all p_i are irreducible and $a_k \neq 0$. If a_k is invertible or irreducible, then we are done. Otherwise, because of Corollary I.5.9, there exist p_{k+1} irreducible and $a_{k+1} \in A$ such that $a_k = p_{k+1} a_{k+1}$. Because each p_i is irreducible, we have strict inclusions:

$$Aa \subsetneqq Aa_1 \subsetneqq \cdots \subsetneqq Aa_k \subsetneqq Aa_{k+1}.$$

Because of Lemma I.5.11, this chain stabilises, that is, there exists k such that a_k is invertible or irreducible. This completes the proof of existence.

(b) *Uniqueness.* Assume that $p_1 \cdots p_m = q_1 \cdots q_n$, with the p_i, q_j irreducible. Then $p_1 \mid q_1 \cdots q_n$. As seen after Theorem I.5.7, there exists q_j such that $p_1 \mid q_j$. Changing the order if necessary, we may assume that $j = 1$ so $p_1 \mid q_1$. But q_1 is irreducible. Hence, there exists u_1 invertible such that $q_1 = u_1 p_1$, that is, p_1 and q_1 are associates. Simplifying, we get $p_2 \cdots p_m = u_1 q_2 \cdots q_n$. We finish by induction. □

For instance, if K is a field, any nonconstant polynomial in $K[t]$ decomposes in the form $p = cp_1 \cdots p_m$, where $c \in K, c \neq 0$, and the p_i are irreducible polynomials, unique if normalised.

The number m of irreducible factors in the unique factorisation of a is sometimes called its ***length***. It follows from Theorem I.5.12 that the length of a is unambiguously defined. Several proofs on principal ideal domains are done by induction on length of their nonzero, noninvertible elements.

Exercises of Section I.5

1. Prove that each of the following is a subalgebra of $\mathbb{Q}$, then that it is a principal ideal domain.
(a) The fractions $\frac{a}{b}$, with b an odd integer.
(b) Given a prime integer p, the fractions $\frac{a}{p^n}$, with $a, n \in \mathbb{Z}$.

2. Let K be a field. Prove that the algebra $K[[t]]$ of formal power series in t, see Example I.3.11(e), is a principal ideal domain, of which any ideal is of the form $\langle t^i \rangle$, for some $i \geq 0$. (Hint: use the order of an element, see Exercise I.3.15).

3. Let a, b be nonzero elements in a principal ideal domain A. Let d be a gcd and m an lcm of a, b. Prove that:
(a) If $a = da', b = db'$, then a', b' are coprime.
(b) There exists $u \in A$ invertible such that $ab = udm$.

4. Let A be a principal ideal domain and $a, b, c \in A$. Prove that:
  (a) $(a,(b,c)) = ((a,b),c)$.
  (b) $(ca, cb) = c(a,b)$.
  (c) $(a + bc, b) = (a,b)$.
  (d) $(a,b) = d, a \mid c, b \mid c$ imply $ab \mid cd$.
  (e) $(a,b) = 1$ if and only if $Aa + Ab = A$, or if and only if $Aa \cap Ab = A(ab)$.
  (f) $(a,b) = 1$ if and only if, for any $c \in A$, $a \mid bc$ implies $a \mid c$.
  (g) If $(a,b) = 1$, then $(a + b, ab) = 1$.
  (h) If $(a,b) = 1$ and $(a,c) = 1$, then $(a, bc) = 1$.
  (i) If $(a,b) = 1$ and $a \mid bc$, then $a \mid c$.
  (j) If $(a,b) = 1$, $a \mid c$ and $b \mid c$, then $ab \mid c$.
  (k) If $(a,b) \neq 1$, then there exists $p \in A$ irreducible such that $p \mid a$ and $p \mid b$.
  (l) If m, n are coprime integers such that $a^m = b^m$ and $a^n = b^n$, then $a = b$.

5. Let A be a principal ideal domain and $a_1, \cdots, a_n \in A$.
  (a) Define what are meant by the gcd and the lcm of the set $\{a_1, \cdots, a_n\}$. Prove that each set containing $n \geq 2$ elements in A has a gcd $(a_1, \cdots, a_n)$ and an lcm $[a_1, \cdots, a_n]$ in A. Moreover, the gcd may be written as $s_1a_1 + \cdots + s_na_n$, for some $s_1, \cdots, s_n \in A$.
  (b) Prove that $[a_1, \cdots, a_n] = a_1 \cdots a_n$ if and only if $(a_i, a_j) = 1$, for $i \neq j$.
  (c) If $a_1, \cdots, a_n$ and $b_1, \cdots, b_n$ are elements of A such that $a_1b_1 = \cdots = a_nb_n = c$, prove that $(a_1, \cdots, a_n)[b_1, \cdots, b_n] = c$.

6. Prove that:
  (a) A normalised polynomial in $\mathbb{C}[t]$ is irreducible if and only if it is of the form $t - \lambda$, for some $\lambda \in \mathbb{C}$.
  (b) If a normalised polynomial in $\mathbb{R}[t]$ is irreducible, then its degree is at most two, and if $p(t) = t^2 + bt + c$, then it is irreducible if and only if $b^2 - 4c < 0$.
  (c) There exist irreducible polynomials in $\mathbb{Q}[t]$ of any positive degree.

7. Let p be a polynomial in $\mathbb{R}[t]$ having a complex root α. Prove that the complex conjugate $\bar{\alpha}$ of α is also a root of p.

8. We claim that the ring $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ of Gaussian integers is a principal ideal domain.
  (a) Prove that the function $\varphi : z \mapsto z\bar{z} = |z|^2$ from $\mathbb{Z}[i]$ to the natural numbers satisfies the following properties:
    (i) $\varphi(z_1z_2) = \varphi(z_1)\varphi(z_2)$, for any $z_1, z_2 \in \mathbb{Z}[i]$.
    (ii) $\varphi(z_1) \leqslant \varphi(z_1z_2)$ for any $z_1, z_2 \neq 0$ in $\mathbb{Z}[i]$.
    (iii) $\varphi(z) = 1$ if and only if z is invertible in $\mathbb{Z}[i]$.
    (iv) If $\varphi(z)$ is a prime integer, then, for any decomposition $z = z'z''$, with $z', z'' \in \mathbb{Z}[i]$, we have z' or z'' invertible (that is, z is irreducible).
  (b) Let $z, z' \in \mathbb{Z}[i]$ with $z' \neq 0$. Then $\frac{z}{z'} = x + yi$, with $x, y \in \mathbb{Q}$. Let $a, b \in \mathbb{Z}$ be such that $|x - a| \leqslant \frac{1}{2}$ and $|y - b| \leqslant \frac{1}{2}$ and set $q = a + bi$. Prove that $\varphi(\frac{z}{z'} - q) < 1$ and deduce that $z = qz' + r$, with $r = 0$ or $\varphi(r) < \varphi(z')$.
  (c) Show that the quotient q and the remainder r obtained in (b) are not necessarily unique.
  (d) Prove that $\mathbb{Z}[i]$ is a principal ideal domain.

9. Prove that $\mathbb{Z}[\sqrt{2}]$ is a principal ideal domain (Hint: use the technique of the previous exercise).

10. Prove that two irreducible elements p, q in a principal ideal domain are associate if and only if $p \mid q$.

11. Prove that the ideal $< 2, t >$ is maximal in $\mathbb{Z}[t]$.

12. Prove that an ideal in a principal ideal domain is maximal if and only if it is prime, see Exercise I.4.18.

13. Let A be a principal ideal domain. Denote by $\ell(a)$ the length of a nonzero, noninvertible element $a \in A$. Assume that $a, b \in A$ are nonzero, noninvertible and such that $a \mid b$. Prove that:

(a) $\ell(a) \leqslant \ell(b)$.
(b) $\ell(a) = \ell(b)$ if and only if a, b are associate.

14. Prove that any algebraically closed field is infinite.

Chapter II
Modules

II.1 Introduction

As seen before, a module is an abelian group whose elements can be multiplied in a reasonable manner by elements of a ring. Modules appear among other places in the representation theory of finite groups: let K be a field and G a finite group, a K-linear representation of G is a group homomorphism from G to the automorphism group of some K-vector space (so elements of G are 'represented' by invertible matrices). One can prove that the classification of K-linear representations of G is equivalent to the classification of modules over the group algebra KG, see Example I.3.11(g). Moreover, it turns out that working with modules is easier and more elegant than working with representations. One defines similarly representations of algebraic structures other than groups, and their study always reduces to that of modules over some algebra. In the present chapter, we study elementary properties of modules and appropriate maps between them. Throughout, K is a commutative ring and our algebras are associative and unitary K-algebras.

II.2 Modules and submodules

II.2.1 Basic definitions and examples

As seen in Definition I.2.12, modules take their scalars inside rings. Because rings and algebras are equivalent concepts, one can talk about modules over an algebra. The definition is the same.

II.2.1 Definition. Let A be a K-algebra, a ***right A-module*** M is an additive abelian group equipped with an external multiplication, or right A-action, $M \times A \to M, (x, a) \mapsto xa,$ for $x \in M, a \in A,$ such that, for any $x, y \in M$ and $a, b \in A,$ we have:

(B1) $x(ab) = (xa)b,$
(B2) $x1 = x,$
(C1) $x(a + b) = xa + xb,$
(C2) $(x + y)a = xa + ya.$

The 1 which appears in (B2) is the multiplicative identity of A. By analogy with vector spaces, elements of M are sometimes called ***vectors*** and those of A ***scalars***.

One defines ***left A-modules*** in the same way. We write M_A or ${}_AM$ to indicate whether M is a right, or left, A-module, respectively. As we have seen, left A-modules can be

An Introduction to Module Theory. Ibrahim Assem and Flávio U. Coelho, Oxford University Press.
 DOI: 10.1093/9780198904939.003.0003

considered as right modules over the opposite algebra A^{op}. This remark allows to work only with right modules, except in cases where a right and a left module structures appear simultaneously.

Any A-module M has a natural K-module structure. Indeed, let $x \in M$ and $\alpha \in K$, then set

$$x\alpha = x(1 \cdot \alpha).$$

It is easily verified that M becomes a right K-module under this operation. Because K is commutative, M is also a left K-module for the same operation (so, we may write $x\alpha = \alpha x$). Thus, if K is a field, an A-module is also a K-vector space.

The arithmetic properties of modules stated in Proposition I.2.14 obviously hold true, but there is an additional one: if M is an A-module, then its multiplication by A is compatible with its multiplication by K. Given $x \in M$, $\alpha \in K$ and $a \in A$, we have

$$x(\alpha a) = (x\alpha)a = (\alpha x)a = \alpha(xa)$$

because of axiom (B1) in Definition II.2.1 and commutativity of K.

We repeat the definition of submodule.

II.2.2 Definition. Let A be a K-algebra and M an A-module. A subset L of M is a ***submodule*** if it is itself an A-module for the same operations as M.

Any module M has at least two submodules, namely the zero submodule 0, which is the one-element set $\{0\}$, and M itself. We restate the criterion of Proposition I.2.16.

II.2.3 Proposition. *Let A be an algebra, M an A-module and L a nonempty subset of M. The following conditions are equivalent:*

(a) *L is a submodule of M.*
(b) *L is an additive subgroup of M and $x \in L, a \in A$ imply $xa \in L$.*
(c) *For any $x, y \in L$ and $a, b \in A$, we have $xa + yb \in L$.*

Proof. Similar to that of Proposition I.2.16 and left to the reader. □

We draw the reader's attention to nonemptiness of L: the empty set satisfies (trivially) condition (c) of the previous proposition, but is not a submodule, because it does not contain the zero element.

Examples of modules and submodules are found in I.2.13 and I.2.17. We give additional ones.

II.2.4 Examples.

(a) Let A be an algebra. The product $(x, a) \mapsto xa$ of $x, a \in A$ endows A itself with a right A-module structure. This module is denoted by A_A. A submodule I of A_A is, according to Proposition II.2.3(b), an additive subgroup of A such that $x \in I, a \in A$ imply $xa \in I$. Such a submodule I is called a ***right ideal*** of A, because its multiplication by elements of A is right stable (a property sometimes expressed by writing $IA \subseteq I$).

Similarly, A has a left A-module structure, given by the product $(a, x) \mapsto ax$ of $a, x \in A$ and denoted by ${}_AA$. A submodule I of ${}_AA$ is an additive subgroup of A such that $x \in I, a \in A$ imply $ax \in I$ (which can be written as $AI \subseteq I$). Then, I is called a ***left ideal*** of A.

It follows from these definitions that a subset I of A is an ideal if and only if it is at the same time a right and a left ideal of A. In this case, it is a submodule of both A_A and ${}_AA$. If A is commutative, then the three notions of ideal, right ideal and left ideal coincide, but this is not the case in non-commutative algebras. For instance, let $A = M_2(K)$ be the 2×2-full matrix algebra, then the set

$$I = \begin{pmatrix} 0 & 0 \\ K & K \end{pmatrix} = \left\{ \begin{pmatrix} 0 & 0 \\ \alpha & \beta \end{pmatrix} : \alpha, \beta \in K \right\}$$

is a right ideal, but not a left ideal of A because A has no nonzero proper ideals, see Example I.3.11(b). Similarly, $I' = \begin{pmatrix} K & 0 \\ K & 0 \end{pmatrix}$ is a left, but not a right ideal of A.

(b) Let A be an algebra and B a subalgebra. Then A admits a right B-module structure defined by $(a, b) \mapsto ab$, for $a \in A, b \in B$, the product being computed in A. For instance, the polynomial algebra $A[t]$ can be considered as a right, and also as a left, A-module.

(c) Let $M_1, \ldots, M_n$ be A-modules, their cartesian product

$$\prod_{i=1}^{n} M_i = M_1 \times \cdots \times M_n = \{(x_1, \cdots, x_n) : x_i \in M_i\}$$

admits an A-module structure, with operations defined componentwise:

$$(x_1, \cdots, x_n) + (y_1, \cdots, y_n) = (x_1 + y_1, \cdots, x_n + y_n)$$
$$(x_1, \cdots, x_n) \cdot a = (x_1 a, \cdots, x_n a)$$

for $x_i, y_i \in M_i, a \in A$. The module $\prod_{i=1}^{n} M_i$ is called the ***product module*** of the M_i.

(d) Let A be an algebra and n a positive integer. Because of Example (c) above, the product $A^n = \{(x_1, \cdots, x_n) : x_i \in A\}$ admits a right, and also a left, A-module structure as product module. But it has another right module structure, this time over the full matrix algebra $M_n(A)$: we give A^n the same additive structure as the product module, but define multiplication of $\boldsymbol{x} = (x_1, \cdots, x_n) \in A^n$ by $\mathbf{a} = [a_{ij}] \in M_n(A)$ as

$$\mathbf{x} \cdot \mathbf{a} = \left(\sum_{i=1}^{n} x_i a_{i1}, \cdots, \sum_{i=1}^{n} x_i a_{in} \right).$$

That is, the product of the row vector $\mathbf{x}$ by the matrix $\mathbf{a}$ is their product as matrices. This is the structure of Exercise I.2.10. Similarly, one defines a left $M_n(A)$-module structure on the set of column vectors with n coordinates in A.

(e) Let A, B be algebras and $\varphi : A \to B$ an algebra morphism. A B-module M_B admits an A-module structure defined by

$$xa = x\varphi(a)$$

for $x \in M, a \in A$. This A-module structure is said to be induced by ***change of scalars*** (because we pass from scalars in B to scalars in A). There are two important particular cases. The first is when A is a subalgebra of B and φ is the inclusion, this is the situation of Example (b) above. The second is when $B = A/I$, for some ideal I of A, and $\varphi : A \to A/I$ is the projection: then, the right A-action on an A/I-module M is given by $xa = x(a + I)$, for $x \in M, a \in A$.

(f) Let K be a field, E a K-vector space and $f : E \to E$ a K-linear map. Define inductively the powers of f as follows: set $f^0 = 1_E$, the identity map, and, for $i \geq 1$, set $f^i = f \circ f^{i-1}$. This data induces a left $K[t]$-module structure on E as follows: addition is the same as in the vector space but multiplication of the polynomial $p = \alpha_0 + \alpha_1 t + \cdots + \alpha_d t^d$ by the vector $v \in E$ is given by

$$p \cdot v = (\alpha_0 1_E + \alpha_1 f + \cdots + \alpha_d f^d)(v) = \alpha_0 v + \alpha_1 f(v) + \cdots + \alpha_d f^d(v).$$

This module is denoted by ${}_fE$ in order to indicate dependence on f. Because $K[t]$ is commutative, E is also a right $K[t]$-module for the same multiplication. However, maps act on the left of elements, so it is more convenient to look at ${}_fE$ as a left module. This module structure will be much used in Chapter V.

We claim that a subset F of E is a $K[t]$-submodule of ${}_fE$ if and only if it is a vector subspace which is moreover ***f-invariant***, that is, such that $f(F) \subseteq F$.

Indeed, suppose that F is an f-invariant subspace of E. Because addition is the same in the module as in the vector space, we need only verify stability of multiplication by an element of $K[t]$. Let $v \in F$. Then, f-invariance of F gives $f(v) \in F$ and induction yields $f^i(v) \in F$, for any integer $i > 0$. Because F is stable under K-linear combinations, then, for any $p \in K[t], v \in F$, we have $p \cdot v \in F$. Therefore, F is a $K[t]$-submodule of ${}_fE$. Conversely, if F is a $K[t]$-submodule, it is an additive subgroup of E and, moreover, for $v \in F, \alpha \in K$, we have $\alpha v \in F$, so that F is a vector subspace of E. Also, for $v \in F$, we have $f(v) = t \cdot v \in F$, and so F is f-invariant. This establishes our claim.

For instance, given $\lambda \in K$, the set $E_\lambda = \{v \in E : f(v) = \lambda v\}$ is an f-invariant subspace of E. It is nonzero if and only if λ is a proper value of f.

(g) Let K be a field and Q a finite quiver. A K-linear ***representation*** V of Q is the following data: